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NON-VOLATILE MEMORY WITH EFFICIENT SETTING OF INITIAL PROGRAM VOLTAGE

Abstract

While programming a first set of memory cells, a non-volatile memory apparatus determines an initial magnitude of a programming signal for a second set of memory cells based on testing during the programming of the first set of memory cells. The testing comprises sensing at a first test voltage level and sensing at a second test voltage level without apply a voltage spike between the two sensing operation. Removing the voltage spike results in a performance increase (i.e. faster programming speed) and as well as a decrease in power usage.

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Background/Summary

BACKGROUND

[0001] The present disclosure relates to non-volatile storage.

[0002] Semiconductor memory is widely used in various electronic devices such as cellular telephones, digital cameras, personal digital assistants, medical electronics, mobile computing devices, servers, solid state drives, non-mobile computing devices and other devices. Semiconductor memory may comprise non-volatile memory or volatile memory. Non-volatile memory allows information to be stored and retained even when the non-volatile memory is not connected to a source of power (e.g., a battery). One example of non-volatile memory is flash memory (e.g., NAND-type and NOR-type flash memory).

[0003] Users of non-volatile memory can program (e.g., write) data to the non-volatile memory and later read that data back. For example, a digital camera may take a photograph and store the photograph in non-volatile memory. Later, a user of the digital camera may view the photograph by having the digital camera read the photograph from the non-volatile memory. When storing data to non-volatile memory, performance is important as users typically do not like to wait for the memory to finish the programming/writing operation.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0004] Like-numbered elements refer to common components in the different figures.

[0005] FIG. **1** is a block diagram depicting one embodiment of a storage system.

[0006] FIG. **2**A is a block diagram of one embodiment of a memory die.

[0007] FIG. **2**B is a block diagram of one embodiment of an integrated memory assembly.

[0008] FIGS. **3**A and **3**B depict different embodiments of integrated memory assemblies.

[0009] FIG. **4** is a perspective view of a portion of one embodiment of a monolithic three dimensional memory structure.

[0010] FIG. **4**A is a block diagram of one embodiment of a memory structure having two planes.

[0011] FIG. **4**B depicts a top view of a portion of one embodiment of a block of memory cells.

[0012] FIG. **4**C depicts a cross sectional view of a portion of one embodiment of a block of memory cells.

[0013] FIG. **4**D depicts a cross sectional view of a portion of one embodiment of a block of memory cells.

[0014] FIG. **4**E is a cross sectional view of one embodiment of a vertical column of memory cells.

[0015] FIG. **4**F is a schematic of a plurality of NAND strings in multiple regions of a same block.

[0016] FIG. **5**A depicts threshold voltage distributions.

[0017] FIG. **5**B depicts threshold voltage distributions.

[0018] FIG. **5**C depicts threshold voltage distributions.

[0019] FIG. **5**D depicts threshold voltage distributions.

[0020] FIG. **6** is a flow chart describing one embodiment of a process for programming non-volatile memory.

[0021] FIGS. **7**, **7**A and **7**B depict voltage signals applied to a selected word line during programming.

[0022] FIG. **8**A depicts two program voltage pulses applied to a selected word line during programming and a verify voltage pulse between the two program voltage pulses.

[0023] FIG. **8**B depicts two program voltage pulses applied to a selected word line during programming and a verify voltage pulse between the two program voltage pulses.

[0024] FIG. **9** depicts multiple threshold voltage distributions representing a group of non-volatile memory cells during the beginning of a programming process.

[0025] FIG. **10** depicts multiple threshold voltage distributions representing a group of non-volatile memory cells during the beginning of a programming process.

[0026] FIG. **11** is a flow chart describing one embodiment of a process for performing a verify process that sets the initial voltage magnitude of a set pf program voltage pulses.

[0027] FIG. **12** is a timing diagram describing certain signals during one embodiment of a programming process.

[0028] FIG. **13** is a timing diagram describing certain signals during one embodiment of a programming process.

[0029] FIG. **14** is a flow chart describing one embodiment of a process for programming non-volatile memory cells.

[0030] FIG. **15** is a flow chart describing one embodiment of a process for programming non-volatile memory cells.

DETAILED DESCRIPTION

[0031] While programming a first set of memory cells, a non-volatile memory apparatus determines an initial magnitude of a programming signal for a second set of memory cells based on testing during the programming of the first set of memory cells. The testing comprises sensing at a first test voltage level and sensing at a second test voltage level without apply a voltage spike between the two sensing operation. Removing the voltage spike results in a performance increase (i.e. faster programming speed) and as well as a decrease in power usage.

[0032] FIG. **1** is a block diagram of one embodiment of a storage system **100** that implements the proposed technology described herein. In one embodiment, storage system **100** is a solid state drive ("SSD"). Storage system **100** can also be a memory card, USB drive or other type of storage system. The proposed technology is not limited to any one type of memory system. Storage system **100** is connected to host **102**, which can be a computer, server, electronic device (e.g., smart phone, tablet or other mobile device), appliance, or another apparatus that uses memory and has data processing capabilities. In some embodiments, host **102** is separate from, but connected to, storage system **100**. In other embodiments, storage system **100** is embedded within host **102**.

[0033] The components of storage system **100** depicted in FIG. **1** are electrical circuits. Storage system **100** includes a memory controller **120** connected to non-volatile memory **130** and local high speed volatile memory **140** (e.g., DRAM). Local high speed volatile memory **140** is used by memory controller **120** to perform certain functions. For example, local high speed volatile memory **140** stores logical to physical address translation tables ("L2P tables").

[0034] Memory controller **120** comprises a host interface **152** that is connected to and in communication with host **102**. In one embodiment, host interface **152** implements a NVM Express (NVMe) over PCI Express (PCIe). Other interfaces can also be used, such as SCSI, SATA, etc. Host interface **152** is also connected to a network-on-chip (NOC) **154**. A NOC is a communication subsystem on an integrated circuit. NOC's can span synchronous and asynchronous clock domains or use unclocked asynchronous logic. NOC technology applies networking theory and methods to on-chip communications and brings notable improvements over conventional bus and crossbar interconnections. NOC improves the scalability of systems on a chip (SoC) and the power efficiency of complex SoCs compared to other designs. The wires and the links of the NOC are shared by many signals. A high level of parallelism is achieved because all links in the NOC can operate simultaneously on different data packets. Therefore, as the complexity of integrated subsystems keep growing, a NOC provides enhanced performance (such as throughput) and scalability in comparison with previous communication architectures (e.g., dedicated point-to-point signal wires, shared buses, or segmented buses with bridges). In other embodiments, NOC **154** can be replaced by a bus. Connected to and in communication with NOC **154** is processor **156**, ECC engine **158**, memory interface **160**, and DRAM controller **164**. DRAM controller **164** is used to operate and communicate with local high speed volatile memory **140** (e.g., DRAM). In other embodiments, local high speed volatile memory **140** can be SRAM or another type of volatile memory.

[0035] ECC engine **158** performs error correction services. For example, ECC engine **158** performs data encoding and decoding, as per the implemented ECC technique. In one embodiment, ECC engine **158** is an electrical circuit programmed by software. For example, ECC engine **158** can be a processor that can be programmed. In other embodiments, ECC engine **158** is a custom and dedicated hardware circuit without any software. In another embodiment, the function of ECC engine **158** is implemented by processor **156**.

[0036] Processor **156** performs the various controller memory operations, such as programming, erasing, reading, and memory management processes. In one embodiment, processor **156** is programmed by firmware. In other embodiments, processor **156** is a custom and dedicated hardware circuit without any software. Processor **156** also implements a translation module, as a software/firmware process or as a dedicated hardware circuit. In many systems, the non-volatile memory is addressed internally to the storage system using physical addresses associated with the one or more memory die. However, the host system will use logical addresses to address the various memory locations. This enables the host to assign data to consecutive logical addresses, while the storage system is free to store the data as it wishes among the locations of the one or more memory die. To implement this system, memory controller **120** (e.g., the translation module) performs address translation between the logical addresses used by the host and the physical addresses used by the memory dies. One example implementation is to maintain tables (i.e., the L2P tables mentioned above) that identify the current translation between logical addresses and physical addresses. An entry in the L2P table may include an identification of a logical address and corresponding physical address. Although logical address to physical address tables (or L2P tables) include the word “tables” they need not literally be tables. Rather, the logical address to physical address tables (or L2P tables) can be any type of data structure. In some examples, the memory space of a storage system is so large that the local memory **140** cannot hold all of the L2P tables. In such a case, the entire set of L2P tables are stored in a memory die **130** and a subset of the L2P tables are cached (L2P cache) in the local high speed volatile memory **140**.

[0037] Memory interface **160** communicates with non-volatile memory **130**. In one embodiment, memory interface provides a Toggle Mode interface. Other interfaces can also be used. In some example implementations, memory interface **160** (or another portion of controller **120**) implements a scheduler and buffer for transmitting data to and receiving data from one or more memory die.

[0038] In one embodiment, non-volatile memory **130** comprises one or more memory die. FIG. **2**A is a functional block diagram of one embodiment of a memory die **200** that comprises non-volatile memory **130**. Each of the one or more memory die of non-volatile memory **130** can be implemented as memory die **200** of FIG. **2**A. The components depicted in FIG. **2**A are electrical circuits. Memory die **200** includes a memory array **202** that can comprises non-volatile memory cells, as described in more detail below. The array terminal lines of memory array **202** include the various layer(s) of word lines organized as rows, and the various layer(s) of bit lines organized as columns. However, other orientations can also be implemented. Memory die **200** includes row control circuitry **220**, whose outputs **208** are connected to respective word lines of the memory array **202**. Row control circuitry **220** receives a group of M row address signals and one or more various control signals from System Control Logic circuit **260**, and typically may include such circuits as row decoders **222**, array terminal drivers **224**, and block select circuitry **226** for both reading and writing (programming) operations. Row control circuitry **220** may also include read/write circuitry. Memory die **200** also includes column control circuitry **210** including sense amplifier(s) **230** whose input/outputs **206** are connected to respective bit lines of the memory array **202**. Although only single block is shown for array **202**, a memory die can include multiple arrays that can be individually accessed. Column control circuitry **210** receives a group of N column address signals and one or more various control signals from System Control Logic **260**, and typically may include such circuits as column decoders **212**, array terminal receivers or driver circuits **214**, block select circuitry **216**, as well as read/write circuitry, and I/O multiplexers.

[0039] System control logic **260** receives data and commands from memory controller **120** and provides output data and status to the host. In some embodiments, the system control logic **260** (which comprises one or more electrical circuits) include state machine **262** that provides die-level control of memory operations. In one embodiment, the state machine **262** is programmable by software. In other embodiments, the state machine **262** does not use software and is completely implemented in hardware (e.g., electrical circuits). In another embodiment, the state machine **262** is replaced by a micro-controller or microprocessor, either on or off the memory chip. System control logic **262** can also include a power control module **264** that controls the power and voltages supplied to the rows and columns of the memory structure **202** during memory operations and may include charge pumps and regulator circuit for creating regulating voltages. System control logic **262** includes storage **366** (e.g., RAM, registers, latches, etc.), which may be used to store parameters for operating the memory array **202**.

[0040] Commands and data are transferred between memory controller **120** and memory die **200** via memory controller interface **268** (also referred to as a “communication interface”). Memory controller interface **268** is an electrical interface for communicating with memory controller **120**. Examples of memory controller interface **268** include a Toggle Mode Interface and an Open NAND Flash Interface (ONFI). Other I/O interfaces can also be used.

[0041] In some embodiments, all the elements of memory die **200**, including the system control logic **260**, can be formed as part of a single die. In other embodiments, some or all of the system control logic **260** can be formed on a different die.

[0042] In one embodiment, memory structure **202** comprises a three-dimensional memory array of non-volatile memory cells in which multiple memory levels are formed above a single substrate, such as a wafer. The memory structure may comprise any type of non-volatile memory that are monolithically formed in one or more physical levels of memory cells having an active area disposed above a silicon (or other type of) substrate. In one example, the non-volatile memory cells comprise vertical NAND strings with charge-trapping layers.

[0043] In another embodiment, memory structure **302** comprises a two-dimensional memory array of non-volatile memory cells. In one example, the non-volatile memory cells are NAND flash memory cells utilizing floating gates. Other types of memory cells (e.g., NOR-type flash memory) can also be used.

[0044] The exact type of memory array architecture or memory cell included in memory structure **202** is not limited to the examples above. Many different types of memory array architectures or memory technologies can be used to form memory structure **202**. No particular non-volatile memory technology is required for purposes of the new claimed embodiments proposed herein. Other examples of suitable technologies for memory cells of the memory structure **202** include ReRAM memories (resistive random access memories), magnetoresistive memory (e.g., MRAM, Spin Transfer Torque MRAM, Spin Orbit Torque MRAM), FeRAM, phase change memory (e.g., PCM), and the like. Examples of suitable technologies for memory cell architectures of the memory structure **202** include two dimensional arrays, three dimensional arrays, cross-point arrays, stacked two dimensional arrays, vertical bit line arrays, and the like.

[0045] One example of a ReRAM cross-point memory includes reversible resistance-switching elements arranged in cross-point arrays accessed by X lines and Y lines (e.g., word lines and bit lines). In another embodiment, the memory cells may include conductive bridge memory elements. A conductive bridge memory element may also be referred to as a programmable metallization cell. A conductive bridge memory element may be used as a state change element based on the physical relocation of ions within a solid electrolyte. In some cases, a conductive bridge memory element may include two solid metal electrodes, one relatively inert (e.g., tungsten) and the other electrochemically active (e.g., silver or copper), with a thin film of the solid electrolyte between the two electrodes. As temperature increases, the mobility of the ions also increases causing the programming threshold for the conductive bridge memory cell to decrease. Thus, the conductive bridge memory element may have a wide range of programming thresholds over temperature.

[0046] Another example is magnetoresistive random access memory (MRAM) that stores data by magnetic storage elements. The elements are formed from two ferromagnetic layers, each of which can hold a magnetization, separated by a thin insulating layer. One of the two layers is a permanent magnet set to a particular polarity; the other layer's magnetization can be changed to match that of an external field to store memory. A memory device is built from a grid of such memory cells. In one embodiment for programming, each memory cell lies between a pair of write lines arranged at right angles to each other, parallel to the cell, one above and one below the cell. When current is passed through them, an induced magnetic field is created. MRAM based memory embodiments will be discussed in more detail below.

[0047] Phase change memory (PCM) exploits the unique behavior of chalcogenide glass. One embodiment uses a GeTe—Sb2Te3 super lattice to achieve non-thermal phase changes by simply changing the co-ordination state of the Germanium atoms with a laser pulse (or light pulse from another source). Therefore, the doses of programming are laser pulses. The memory cells can be inhibited by blocking the memory cells from receiving the light. In other PCM embodiments, the memory cells are programmed by current pulses. Note that the use of “pulse” in this document does not require a square pulse but includes a (continuous or non-continuous) vibration or burst of sound, current, voltage light, or another wave. These memory elements within the individual selectable memory cells, or bits, may include a further series element that is a selector, such as an ovonic threshold switch or metal insulator substrate.

[0048] A person of ordinary skill in the art will recognize that the technology described herein is not limited to a single specific memory structure, memory construction or material composition, but covers many relevant memory structures within the spirit and scope of the technology as described herein and as understood by one of ordinary skill in the art.

[0049] The elements of FIG. **2**A can be grouped into two parts: (1) memory structure **202** and (2) peripheral circuitry, which includes all of the other components depicted in FIG. **2**A. An important characteristic of a memory circuit is its capacity, which can be increased by increasing the area of the memory die of storage system **100** that is given over to the memory structure **202**; however, this reduces the area of the memory die available for the peripheral circuitry. This can place quite severe restrictions on these elements of the peripheral circuitry. For example, the need to fit sense amplifier circuits within the available area can be a significant restriction on sense amplifier design architectures. With respect to the system control logic **260**, reduced availability of area can limit the available functionalities that can be implemented on-chip. Consequently, a basic trade-off in the design of a memory die for the storage system **100** is the amount of area to devote to the memory structure **202** and the amount of area to devote to the peripheral circuitry.

[0050] Another area in which the memory structure **202** and the peripheral circuitry are often at odds is in the processing involved in forming these regions, since these regions often involve differing processing technologies and the trade-off in having differing technologies on a single die. For example, when the memory structure **202** is NAND flash, this is an NMOS structure, while the peripheral circuitry is often CMOS based. For example, elements such sense amplifier circuits, charge pumps, logic elements in a state machine, and other peripheral circuitry in system control logic **260** often employ PMOS devices. Processing operations for manufacturing a CMOS die will differ in many aspects from the processing operations optimized for an NMOS flash NAND memory or other memory cell technologies.

[0051] To improve upon these limitations, embodiments described below can separate the elements of FIG. **2**A onto separately formed dies that are then bonded together. More specifically, the memory structure **202** can be formed on one die (referred to as the memory die) and some or all of the peripheral circuitry elements, including one or more control circuits, can be formed on a separate die (referred to as the control die). For example, a memory die can be formed of just the memory elements, such as the array of memory cells of flash NAND memory, MRAM memory, PCM memory, ReRAM memory, or other memory type. Some or all of the peripheral circuitry, even including elements such as decoders and sense amplifiers, can then be moved on to a separate control die. This allows each of the memory die to be optimized individually according to its technology. For example, a NAND memory die can be optimized for an NMOS based memory array structure, without worrying about the CMOS elements that have now been moved onto a control die that can be optimized for CMOS processing. This allows more space for the peripheral elements, which can now incorporate additional capabilities that could not be readily incorporated were they restricted to the margins of the same die holding the memory cell array. The two die can then be bonded together in a bonded multi-die memory circuit, with the array on the one die connected to the periphery elements on the other die. Although the following will focus on a bonded memory circuit of one memory die and one control die, other embodiments can use more die, such as two memory die and one control die, for example.

[0052] FIG. **2**B shows an alternative arrangement to that of FIG. **2**A which may be implemented using wafer-to-wafer bonding to provide a bonded die pair. FIG. **2**B depicts a functional block diagram of one embodiment of an integrated memory assembly **207**. One or more integrated memory assemblies **207** may be used to implement the non-volatile memory **130** of storage system **100**. The integrated memory assembly **207** includes two types of semiconductor die (or more succinctly, “die”). Memory die **201** includes memory structure **202**. Memory structure **202** includes non-volatile memory cells. Control die **211** includes control circuitry **260**, **210**, and **220** (as described above). In some embodiments, control die **211** is configured to connect to the memory structure **202** in the memory die **201**. In some embodiments, the memory die **201** and the control die **211** are bonded together.

[0053] FIG. **2**B shows an example of the peripheral circuitry, including control circuits, formed in a peripheral circuit or control die **211** coupled to memory structure **202** formed in memory die **201**. Common components are labelled similarly to FIG. **2**A. System control logic **260**, row control circuitry **220**, and column control circuitry **210** are located in control die **211**. In some embodiments, all or a portion of the column control circuitry **210** and all or a portion of the row control circuitry **220** are located on the memory die **201**. In some embodiments, some of the circuitry in the system control logic **260** is located on the on the memory die **201**.

[0054] System control logic **260**, row control circuitry **220**, and column control circuitry **210** may be formed by a common process (e.g., CMOS process), so that adding elements and functionalities, such as ECC, more typically found on a memory controller **120** may require few or no additional process steps (i.e., the same process steps used to fabricate controller **120** may also be used to fabricate system control logic **260**, row control circuitry **220**, and column control circuitry **210**). Thus, while moving such circuits from a die such as memory **2** die **201** may reduce the number of steps needed to fabricate such a die, adding such circuits to a die such as control die **211** may not require many additional process steps. The control die **211** could also be referred to as a CMOS die, due to the use of CMOS technology to implement some or all of control circuitry **260**, **210**, **220**.

[0055] FIG. **2**B shows column control circuitry **210** including sense amplifier(s) **230** on the control die **211** coupled to memory structure **202** on the memory die **201** through electrical paths **206**. For example, electrical paths **206** may provide electrical connection between column decoder **212**, driver circuitry **214**, and block select **216** and bit lines of memory structure **202**. Electrical paths may extend from column control circuitry **210** in control die **211** through pads on control die **211** that are bonded to corresponding pads of the memory die **201**, which are connected to bit lines of memory structure **202**. Each bit line of memory structure **202** may have a corresponding electrical path in electrical paths **206**, including a pair of bond pads, which connects to column control circuitry **210**. Similarly, row control circuitry **220**, including row decoder **222**, array drivers **224**, and block select **226** are coupled to memory structure **202** through electrical paths **208**. Each of electrical path **208** may correspond to a word line, dummy word line, or select gate line. Additional electrical paths may also be provided between control die **211** and memory die **201**.

[0056] For purposes of this document, the phrases “a control circuit” or “one or more control circuits” can include any one of or any combination of memory controller **120**, state machine **262**, all or a portion of system control logic **260**, all or a portion of row control circuitry **220**, all or a portion of column control circuitry **210**, a microcontroller, a microprocessor, and/or other similar functioned circuits. The control circuit can include hardware only or a combination of hardware and software (including firmware). For example, a controller programmed by firmware to perform the functions described herein is one example of a control circuit. A control circuit can include a processor, FGA, ASIC, integrated circuit, or other type of circuit.
[0057] In some embodiments, there is more than one control die **211** and more than one memory die **201** in an integrated memory assembly **207**. In some embodiments, the integrated memory assembly **207** includes a stack of multiple control die **211** and multiple memory die **201**. FIG. **3**A depicts a side view of an embodiment of an integrated memory assembly **207** stacked on a substrate **271** (e.g., a stack comprising control dies **211** and memory dies **201**). The integrated memory assembly **207** has three control dies **211** and three memory dies **201**. In some embodiments, there are more than three memory dies **201** and more than three control die **211**.
[0058] Each control die **211** is affixed (e.g., bonded) to at least one of the memory dies **201**. Some of the bond pads **282**/**284** are depicted. There may be many more bond pads. A space between two dies **201**, **211** that are bonded together is filled with a solid layer **280**, which may be formed from epoxy or other resin or polymer. This solid layer **280** protects the electrical connections between the dies **201**, **211**, and further secures the dies together. Various materials may be used as solid layer **280**, but in embodiments, it may be Hysol epoxy resin from Henkel Corp., having offices in California, USA.
[0059] The integrated memory assembly **207** may for example be stacked with a stepped offset, leaving the bond pads at each level uncovered and accessible from above. Wire bonds **270** connected to the bond pads connect the control die **211** to the substrate **271**. A number of such wire bonds may be formed across the width of each control die **211** (i.e., into the page of FIG. **3**A).
[0060] A memory die through silicon via (TSV) **276** may be used to route signals through a memory die **201**. A control die through silicon via (TSV) **278** may be used to route signals through a control die **211**. The TSVs **276**, **278** may be formed before, during or after formation of the integrated circuits in the semiconductor dies **201**, **211**. The TSVs may be formed by etching holes through the wafers. The holes may then be lined with a barrier against metal diffusion. The barrier layer may in turn be lined with a seed layer, and the seed layer may be plated with an electrical conductor such as copper, although other suitable materials such as aluminum, tin, nickel, gold, doped polysilicon, and alloys or combinations thereof may be used.
[0061] Solder balls **272** may optionally be affixed to contact pads **274** on a lower surface of substrate **271**. The solder balls **272** may be used to couple the integrated memory assembly **207** electrically and mechanically to a host device such as a printed circuit board. Solder balls **272** may be omitted where the integrated memory assembly **207** is to be used as an LGA package. The solder balls **272** may form a part of the interface between integrated memory assembly **207** and memory controller **120**.
[0062] FIG. **3**B depicts a side view of another embodiment of an integrated memory assembly **207** stacked on a substrate **271**. The integrated memory assembly **207** of FIG. **3**B has three control die **211** and three memory die **201**. In some embodiments, there are many more than three memory dies **201** and many more than three control dies **211**. In this example, each control die **211** is bonded to at least one memory die **201**. Optionally, a control die **211** may be bonded to two or more memory die **201**.
[0063] Some of the bond pads **282**, **284** are depicted. There may be many more bond pads. A space between two dies **201**, **211** that are bonded together is filled with a solid layer **280**, which may be formed from epoxy or other resin or polymer. In contrast to the example in FIG. **3**A, the integrated memory assembly **207** in FIG. **3**B does not have a stepped offset. A memory die through silicon via

(TSV) **276** may be used to route signals through a memory die **201**. A control die through silicon via (TSV) **278** may be used to route signals through a control die **211**.

[0064] Solder balls **272** may optionally be affixed to contact pads **274** on a lower surface of substrate **271**. The solder balls **272** may be used to couple the integrated memory assembly **207** electrically and mechanically to a host device such as a printed circuit board. Solder balls **272** may be omitted where the integrated memory assembly **207** is to be used as an LGA package.

[0065] As has been briefly discussed above, the control die **211** and the memory die **201** may be bonded together. Bond pads on each die **201**, **211** may be used to bond the two dies together. In some embodiments, the bond pads are bonded directly to each other, without solder or other added material, in a so-called Cu-to-Cu bonding process. In a Cu-to-Cu bonding process, the bond pads are controlled to be highly planar and formed in a highly controlled environment largely devoid of ambient particulates that might otherwise settle on a bond pad and prevent a close bond. Under such properly controlled conditions, the bond pads are aligned and pressed against each other to form a mutual bond based on surface tension. Such bonds may be formed at room temperature, though heat may also be applied. In embodiments using Cu-to-Cu bonding, the bond pads may be about 5 μm square and spaced from each other with a pitch of 5 μm to 5 μm. While this process is referred to herein as Cu-to-Cu bonding, this term may also apply even where the bond pads are formed of materials other than Cu.

[0066] When the area of bond pads is small, it may be difficult to bond the semiconductor dies together. The size of, and pitch between, bond pads may be further reduced by providing a film layer on the surfaces of the semiconductor dies including the bond pads. The film layer is provided around the bond pads. When the dies are brought together, the bond pads may bond to each other, and the film layers on the respective dies may bond to each other. Such a bonding technique may be referred to as hybrid bonding. In embodiments using hybrid bonding, the bond pads may be about 5 μm square and spaced from each other with a pitch of 1 μm to 5 μm. Bonding techniques may be used providing bond pads with even smaller (or greater) sizes and pitches.

[0067] Some embodiments may include a film on surface of the dies **201**, **211**. Where no such film is initially provided, a space between the dies may be under filled with an epoxy or other resin or polymer. The under-fill material may be applied as a liquid which then hardens into a solid layer. This under-fill step protects the electrical connections between the dies **201**, **211**, and further secures the dies together. Various materials may be used as under-fill material, but in embodiments, it may be Hysol epoxy resin from Henkel Corp., having offices in California, USA.

[0068] FIG. **4** is a perspective view of a portion of one example embodiment of a monolithic three dimensional memory array/structure that can comprise memory structure **202**, which includes a plurality non-volatile memory cells arranged as vertical NAND strings. For example, FIG. **4** shows a portion **400** of one block of memory. The structure depicted includes a set of bit lines BL positioned above a stack **401** of alternating dielectric layers and conductive layers. For example purposes, one of the dielectric layers is marked as D and one of the conductive layers (also called word line layers) is marked as W. The number of alternating dielectric layers and conductive layers can vary based on specific implementation requirements. As will be explained below, in one embodiment the alternating dielectric layers and conductive layers are divided into four or five (or a different number of) regions by isolation regions IR. FIG. **4** shows one isolation region IR separating two regions. Below the alternating dielectric layers and word line layers is a source line layer SL. Memory holes are formed in the stack of alternating dielectric layers and conductive layers. For example, one of the memory holes is marked as MH. Note that in FIG. **4**, the dielectric layers are depicted as see-through so that the reader can see the memory holes positioned in the stack of alternating dielectric layers and conductive layers. In one embodiment, NAND strings are formed by filling the memory hole with materials including a charge-trapping material to create a vertical column of memory cells. Each memory cell can store one or more bits of data. Thus, the non-volatile memory cells are arranged in memory holes. More details of the three dimensional

monolithic memory array that comprises memory structure **202** is provided below.
[0069] FIG. **4**A is a block diagram explaining one example organization of memory structure **202**, which is divided into four planes **402**, **403**, **404** and **405**. Each plane is then divided into M blocks. In one example, each plane has about 2000 blocks. However, different numbers of blocks and planes can also be used. In one embodiment, a block of memory cells is a unit of erase. That is, all memory cells of a block are erased together. In other embodiments, blocks can be divided into sub-blocks and the sub-blocks can be the unit of erase. Memory cells can also be grouped into blocks for other reasons, such as to organize the memory structure to enable the signaling and selection circuits. In some embodiments, a block represents a groups of connected memory cells as the memory cells of a block share a common set of word lines. For example, the word lines for a block are all connected to all of the vertical NAND strings for that block. Although FIG. **4**A shows four planes, more or less than four planes can be implemented. In some embodiments, memory structure **202** includes eight planes.
[0070] FIGS. **4**B-**4**G depict an example three dimensional ("3D") NAND structure that corresponds to the structure of FIG. **4** and can be used to implement memory structure **202** of FIGS. **2**A and **2**B. FIG. **4**B is a block diagram depicting a top view of a portion **406** of Block **2** of plane **402**. As can be seen from FIG. **4**B, the block depicted in FIG. **4**B extends in the direction of **432**. In one embodiment, the memory array has many layers; however, FIG. **4**B only shows the top layer.
[0071] FIG. **4**B depicts a plurality of circles that represent the memory holes, which are also referred to as vertical columns. Each of the memory holes/vertical columns include multiple select transistors (also referred to as a select gate or selection gate) and multiple memory cells. In one embodiment, each memory hole/vertical column implements a NAND string. For example, FIG. **4**B labels a subset of the memory holes/vertical columns/NAND strings **432**, **436**, **446**. **456**, **462**, **466**, **472**, **474** and **476**.
[0072] FIG. **4**B also depicts a set of bit lines **415**, including bit lines **411**, **412**, **413**, **414**, . . . **419**. FIG. **4**B shows twenty four bit lines because only a portion of the block is depicted. It is contemplated that more than twenty four bit lines connected to memory holes/vertical columns of the block. Each of the circles representing memory holes/vertical columns has an "x" to indicate its connection to one bit line. For example, bit line **411** is connected to memory holes/vertical columns **436**, **446**, **456**, **466** and **476**.
[0073] The block depicted in FIG. **4**B includes a set of isolation regions **482**, **484**, **486** and **488**, which are formed of SiO.sub.2; however, other dielectric materials can also be used. Isolation regions **482**, **484**, **486** and **488** serve to divide the top layers of the block into five regions; for example, the top layer depicted in FIG. **4**B is divided into regions **430**, **440**, **450**, **460** and **470**. In one embodiment, the isolation regions only divide the layers used to implement select gates so that NAND strings in different regions can be independently selected. In one example implementation, a bit line connects to one memory hole/vertical column/NAND string in each of regions **430**, **440**, **450**, **460** and **470**. In that implementation, each block has twenty four rows of active columns and each bit line connects to five rows in each block. In one embodiment, all of the five memory holes/vertical columns/NAND strings connected to a common bit line are connected to the same set of word lines; therefore, the system uses the drain side selection lines to choose one (or another subset) of the five to be subjected to a memory operation (program, verify, read, and/or erase).
[0074] FIG. **4**B also shows Line Interconnects LI, which are metal connections to the source line SL from above the memory array. Line Interconnects LI are positioned adjacent regions **430** and **470**.
[0075] Although FIG. **4**B shows each region **430**, **440**, **450**, **460** and **470** having four rows of memory holes/vertical columns, five regions and twenty four rows of memory holes/vertical columns in a block, those exact numbers are an example implementation. Other embodiments may include more or less regions per block, more or less rows of memory holes/vertical columns per

region and more or less rows of vertical columns per block. FIG. **4**B also shows the memory holes/vertical columns being staggered. In other embodiments, different patterns of staggering can be used. In some embodiments, the memory holes/vertical columns are not staggered.

[0076] FIG. **4**C depicts a portion of one embodiment of a three dimensional memory structure **202** showing a cross-sectional view along line AA of FIG. **4**B. This cross sectional view cuts through memory holes/vertical columns (NAND strings) **472** and **474** of region **470** (see FIG. **4**B). The structure of FIG. **4**C includes two drain side select layers SGD**0** and SGD; teo source side select layers SGS**0** and SGS**1**; two drain side GIDL generation transistor layers SGDT**0** and SGDT**1**; two source side GIDL generation transistor layers SGSB**0** and SGSB**1**; two drain side dummy word line layers DD**0** and DD**1**; two source side dummy word line layers DS**0** and DS**1**; dummy word line layers DU and DL; one hundred and sixty two word line layers WL**0**-WL**161** for connecting to data memory cells, and dielectric layers DL. Other embodiments can implement more or less than the numbers described above for FIG. **4**C. In one embodiment, SGD**0** and SGD**1** are connected together; and SGS**0** and SGS**1** are connected together. In other embodiments, more or less number of SGDs (greater or lesser than two) are connected together, and more or less number of SGSs (greater or lesser than two) connected together.

[0077] In one embodiment, erasing the memory cells is performed using gate induced drain leakage (GIDL), which includes generating charge carriers at the GIDL generation transistors such that the carriers get injected into the charge trapping layers of the NAND strings to change threshold voltage of the memory cells. FIG. **4**C shows two GIDL generation transistors at each end of the NAND string; however, in other embodiments there are more or less than three. Embodiments that use GIDL at both sides of the NAND string may have GIDL generation transistors at both sides. Embodiments that use GIDL at only the drain side of the NAND string may have GIDL generation transistors only at the drain side. Embodiments that use GIDL at only the source side of the NAND string may have GIDL generation transistors only at the source side.

[0078] FIG. **4**C shows two GIDL generation transistors at each end of the NAND string. It is likely that charge carriers are only generated by GIDL at one of the two GIDL generation transistors at each end of the NAND string. Based on process variances during manufacturing, it is likely that one of the two GIDL generation transistors at an end of the NAND string is best suited for GIDL. For example, the GIDL generation transistors have an abrupt pn junction to generate the charge carriers for GIDL and, during fabrication, a phosphorous diffusion is performed at the polysilicon channel of the GIDL generation transistors. In some cases, the GIDL generation transistor with the shallowest phosphorous diffusion is the GIDL generation transistor that generates the charge carriers during erase. However, in some embodiments charge carriers can be generated by GIDL at multiple GIDL generation transistors at a particular side of the NAND string.

[0079] Memory holes/Vertical columns **472** and **474** are depicted protruding through the drain side select layers, source side select layers, dummy word line layers, GIDL generation transistor layers and word line layers. In one embodiment, each memory hole/vertical column comprises a vertical NAND string. Below the memory holes/vertical columns and the layers listed below is substrate **453**, an insulating film **454** on the substrate, and source line SL. The NAND string of memory hole/vertical column **472** has a source end at a bottom of the stack and a drain end at a top of the stack. As in agreement with FIG. **4**B, FIG. **4**C show vertical memory hole/column **472** connected to bit line **414** via connector **417**.

[0080] For ease of reference, drain side select layers; source side select layers, dummy word line layers, GIDL generation transistor layers and data word line layers collectively are referred to as the conductive layers. In one embodiment, the conductive layers are made from a combination of TiN and Tungsten. In other embodiments, other materials can be used to form the conductive layers, such as doped polysilicon, metal such as Tungsten, metal silicide, such as nickel silicide, tungsten silicide, aluminum silicide or the combination thereof. In some embodiments, different conductive layers can be formed from different materials. Between conductive layers are dielectric layers DL. In one embodiment, the dielectric layers are made from SiO.sub.2. In other embodiments, other dielectric materials can be used to form the dielectric layers.

[0081] The non-volatile memory cells are formed along memory holes/vertical columns which extend through alternating conductive and dielectric layers in the stack. In one embodiment, the memory cells are arranged in NAND strings. The word line layers WL**0**-W**161** connect to memory cells (also called data memory cells). Dummy word line layers connect to dummy memory cells. A dummy memory cell does not store and is not eligible to store host data (data provided from the host, such as data from a user of the host), while a data memory cell is eligible to store host data. In some embodiments, data memory cells and dummy memory cells may have a same structure. Drain side select layers SGD**0** and SGD**1** are used to electrically connect and disconnect NAND strings from bit lines. Source side select layers SGS**0** and SGS**1** are used to electrically connect and disconnect NAND strings from the source line SL.

[0082] FIG. **4**C shows that the memory array is implemented as a two tier architecture, with the tiers separated by a Joint area. In one embodiment it is expensive and/or challenging to etch so many word line layers intermixed with dielectric layers. To case this burden, one embodiment includes laying down a first stack of word line layers (e.g., WL**0**-WL**80**) alternating with dielectric layers, laying down the Joint area, and laying down a second stack of word line layers (e.g., WL**81**-WL**161**) alternating with dielectric layers. The Joint area are positioned between the first stack and the second stack. In one embodiment, the Joint areas are made from the same materials as the word line layers. In other embodiments, there can no Joint area or there can be multiple Joint areas.

[0083] FIG. **4**D depicts a portion of one embodiment of a three dimensional memory structure **202** showing a cross-sectional view along line BB of FIG. **4**B. This cross sectional view cuts through memory holes/vertical columns (NAND strings) **432** and **434** of region **430** (see FIG. **4**B). FIG. **4**D shows the same alternating conductive and dielectric layers as FIG. **4**C. FIG. **4**D also shows isolation region **482**. Isolation regions **482**, **484**, **486** and **488**) occupy space that would have been used for a portion of the memory holes/vertical columns/NAND stings. For example, isolation region **482** occupies space that would have been used for a portion of memory hole/vertical column **434**. More specifically, a portion (e.g., half the diameter) of vertical column **434** has been removed in layers SGDT**0**, SGDT**1**, SGD**0**, and SGD**1** to accommodate isolation region **482**. Thus, while most of the vertical column **434** is cylindrical (with a circular cross section), the portion of vertical column **434** in layers SGDT**0**, SGDT**1**, SGD**0**, and SGD**1** has a semi-circular cross section. In one embodiment, after the stack of alternating conductive and dielectric layers is formed, the stack is etched to create space for the isolation region and that space is then filled in with SiO.sub.2. This structure allows for separate control of SGDT**0**, SGDT**1**, SGD**0**, and SGD**1** for regions **430**, **440**, **450**, **460**, and **470**.

[0084] FIG. **4**E depicts a cross sectional view of region **429** of FIG. **4**C that includes a portion of memory hole/vertical column **472**. In one embodiment, the memory holes/vertical columns are round; however, in other embodiments other shapes can be used. In one embodiment, memory hole/vertical column **472** includes an inner core layer **490** that is made of a dielectric, such as SiO.sub.2. Other materials can also be used. Surrounding inner core **490** is polysilicon channel **491**. Materials other than polysilicon can also be used. Note that it is the channel **491** that connects to the bit line and the source line. Surrounding channel **491** is a tunneling dielectric **492**. In one embodiment, tunneling dielectric **492** has an ONO structure. Surrounding tunneling dielectric **492** is charge trapping layer **493**, such as (for example) Silicon Nitride. Other memory materials and structures can also be used. The technology described herein is not limited to any particular material or structure.

[0085] FIG. **4**E depicts dielectric layers DL as well as word line layers WL**160**, WL**159**, WL**158**, WL**157**, and WL**156**. Each of the word line layers includes a word line region **496** surrounded by an aluminum oxide layer **497**, which is surrounded by a blocking oxide layer **498**. In other embodiments, the blocking oxide layer can be a vertical layer parallel and adjacent to charge trapping layer **493**. The physical interaction of the word line layers with the vertical column forms the memory cells. Thus, a memory cell, in one embodiment, comprises channel **491**, tunneling dielectric **492**, charge trapping layer **493**, blocking oxide layer **498**, aluminum oxide layer **497** and word line region **496**. For example, word line layer WL**160** and a portion of memory hole/vertical column **472** comprise a memory cell MC**1**. Word line layer WL**159** and a portion of memory hole/vertical column **472** comprise a memory cell MC**2**. Word line layer WL**158** and a portion of memory hole/vertical column **472** comprise a memory cell MC**3**. Word line layer WL**157** and a portion of memory hole/vertical column **472** comprise a memory cell MC**4**. Word line layer WL**156** and a portion of memory hole/vertical column **472** comprise a memory cell MC**5**. In other architectures, a memory cell may have a different structure; however, the memory cell would still be the storage unit.

[0086] When a memory cell is programmed, electrons are stored in a portion of the charge trapping layer **493** which is associated with (e.g. in) the memory cell. These electrons are drawn into the charge trapping layer **493** from the channel **491**, through the tunneling dielectric **492**, in response to an appropriate voltage on word line region **496**. The threshold voltage (Vth) of a memory cell is increased in proportion to the amount of stored charge. In one embodiment, the programming is achieved through Fowler-Nordheim tunneling of the electrons into the charge trapping layer. During an erase operation, the electrons return to the channel or holes are injected into the charge trapping layer to recombine with electrons. In one embodiment, erasing is achieved using hole injection into the charge trapping layer via a physical mechanism such as GIDL.

[0087] FIG. **4**F is a schematic diagram of a portion of the three dimensional memory array **202** depicted in in FIGS. **4-4**E. FIG. **4**F shows physical data word lines WL**0**-WL**161** running across the entire block. The structure of FIG. **4**F corresponds to a portion **406** in Block **2** of FIG. **4**A, including bit line **411**. Within the block, in one embodiment, each bit line is connected to five NAND strings, one in each region of regions **430**, **440**, **450**, **460**, **470**. Thus, FIG. **4**F shows bit line **411** connected to NAND string NS**0** (which corresponds to memory hole/vertical column **436** of region **430**), NAND string NS**1** (which corresponds to memory hole/vertical column **446** of region **440**), NAND string NS**2** (which corresponds to vertical column **456** of region **450**), NAND string NS**3** (which corresponds to memory hole/vertical column **466** of region **460**), and NAND string NS**4** (which corresponds to memory hole/vertical column **476** of region **470**).

[0088] Drain side select line/layer SGD**0** is separated by isolation regions isolation regions **482**, **484**, **486** and **488** to form SGD**0**-*s***0**, SGD**0**-*s***1**, SGD**0**-*s***2**, SGD**0**-*s***3** and SGD**0**-*s***4** in order to separately connect to and independently control regions **430**, **440**, **450**, **460**, **470**. Similarly, drain side select line/layer SGD**1** is separated by isolation regions **482**, **484**, **486** and **488** to form SGD**1**-*s***0**, SGD**1**-*s***1**, SGD**1**-*s***2**, SGD**1**-*s***3** and SGD**1**-*s***4** in order to separately connect to and independently control regions **430**, **440**, **450**, **460**, **470**; drain side GIDL generation transistor control line/layer SGDT**0** is separated by isolation regions **482**, **484**, **486** and **488** to form SGDT**0**-*s***0**, SGDT**0**-*s***1**, SGDT**0**-*s***2**, SGDT**0**-*s***3** and SGDT**0**-*s***4** in order to separately connect to and independently control regions **430**, **440**, **450**, **460**, **470**; drain side GIDL generation transistor control line/layer SGDT**1** is separated by isolation regions **482**, **484**, **486** and **488** to form SGDT**1**-*s***0**, SGDT**1**-*s***1**, SGDT**1**-*s***2**, SGDT**1**-*s***3** and SGDT**1**-*s***4** in order to separately connect to and independently control regions **430**, **440**, **450**, **460**, **470**.

[0089] FIG. **4**F only shows NAND strings connected to bit line **411**. However, a full schematic of the block would show every bit line and five vertical NAND strings (that are in separate regions) connected to each bit line.

[0090] Although the example memories of FIGS. **4-4**F are three dimensional memory structure that includes vertical NAND strings with charge-trapping material, other (2D and 3D) memory structures can also be used with the technology described herein.

[0091] The memory systems discussed above can be erased, programmed and read. At the end of a successful programming process, the threshold voltages of the memory cells should be within one

or more distributions of threshold voltages for programmed memory cells or within a distribution of threshold voltages for erased memory cells, as appropriate. FIG. **5**A is a graph of threshold voltage versus number of memory cells, and illustrates example threshold voltage distributions for the memory array when each memory cell stores one bit of data per memory cell. Memory cells that store one bit of data per memory cell data are referred to as single level cells (“SLC”). The data stored in SLC memory cells is referred to as SLC data; therefore, SLC data comprises one bit per memory cell. Data stored as one bit per memory cell is SLC data. FIG. **5**A shows two threshold voltage distributions: E and P. Threshold voltage distribution E corresponds to an erased data state. Threshold voltage distribution P corresponds to a programmed data state. Memory cells that have threshold voltages in threshold voltage distribution E are, therefore, in the erased data state (e.g., they are erased). Memory cells that have threshold voltages in threshold voltage distribution P are, therefore, in the programmed data state (e.g., they are programmed). In one embodiment, erased memory cells store data “1” and programmed memory cells store data “0.” FIG. **5**A depicts read reference voltage Vr. By testing (e.g., performing one or more sense operations) whether the threshold voltage of a given memory cell is above or below Vr, the system can determine a memory cells is erased (state E) or programmed (state P). FIG. **5**A also depicts verify reference voltage Vv. In some embodiments, when programming memory cells to data state P, the system will test whether those memory cells have a threshold voltage greater than or equal to Vv.
[0092] FIGS. **5**B-D illustrate example threshold voltage distributions for the memory array when each memory cell stores multiple bit per memory cell data. Memory cells that store multiple bit per memory cell data are referred to as multi-level cells (“MLC”). The data stored in MLC memory cells is referred to as MLC data; therefore, MLC data comprises multiple bits per memory cell. Data stored as multiple bits of data per memory cell is MLC data. In the example embodiment of FIG. **5**B, each memory cell stores two bits of data. Other embodiments may use other data capacities per memory cell (e.g., such as three, four, or five bits of data per memory cell).
[0093] FIG. **5**B shows a first threshold voltage distribution E for erased memory cells. Three threshold voltage distributions A, B and C for programmed memory cells are also depicted. In one embodiment, the threshold voltages in the distribution E are negative and the threshold voltages in distributions A, B and C are positive. Each distinct threshold voltage distribution of FIG. **5**B corresponds to predetermined values for the set of data bits. In one embodiment, each bit of data of the two bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP) and an upper page (UP). In other embodiments, all bits of data stored in a memory cell are in a common logical page. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. Table 1 provides an example encoding scheme.
TABLE-US-00001 TABLE 1 E A B C LP 1 0 0 1 UP 1 1 0 0
[0094] In one embodiment, known as full sequence programming, memory cells can be programmed from the erased data state E directly to any of the programmed data states A, B or C using the process of FIG. **6** (discussed below). For example, a population of memory cells to be programmed may first be erased so that all memory cells in the population are in erased data state E. Then, a programming process is used to program memory cells directly into data states A, B, and/or C. For example, while some memory cells are being programmed from data state E to data state A, other memory cells are being programmed from data state E to data state B and/or from data state E to data state C. The arrows of FIG. **5**B represent the full sequence programming. In some embodiments, data states A-C can overlap, with memory controller **120** (or control die **211**) relying on error correction to identify the correct data being stored.
[0095] FIG. **5**C depicts example threshold voltage distributions for memory cells where each memory cell stores three bits of data per memory cells (which is another example of MLC data). FIG. **5**C shows eight threshold voltage distributions, corresponding to eight data states. The first threshold voltage distribution (data state) Er represents memory cells that are erased. The other

seven threshold voltage distributions (data states) A-G represent memory cells that are programmed and, therefore, are also called programmed states. Each threshold voltage distribution (data state) corresponds to predetermined values for the set of data bits. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. In one embodiment, data values are assigned to the threshold voltage ranges using a Gray code assignment so that if the threshold voltage of a memory erroneously shifts to its neighboring physical state, only one bit will be affected. Table 2 provides an example of an encoding scheme for embodiments in which each bit of data of the three bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP), middle page (MP) and an upper page (UP).

TABLE-US-00002 TABLE 2 Er A B C D E F G UP 1 1 1 0 0 0 0 1 MP 1 1 0 0 1 1 0 0 LP 1 0 0 0 0 1 1 1

[0096] FIG. **5**C shows seven read reference voltages, VrA, VrB, VIC, VrD, VrE, VrF, and VrG for reading data from memory cells. By testing (e.g., performing sense operations) whether the threshold voltage of a given memory cell is above or below the seven read reference voltages, the system can determine what data state (i.e., A, B, C, D, . . .) a memory cell is in.

[0097] FIG. **5**C also shows seven verify reference voltages, VvA, VvB, VvC, VvD, VvE, VvF, and VvG. In some embodiments, when programming memory cells to data state A, the system will test whether those memory cells have a threshold voltage greater than or equal to VvA. When programming memory cells to data state B, the system will test whether the memory cells have threshold voltages greater than or equal to VvB. When programming memory cells to data state C, the system will determine whether memory cells have their threshold voltage greater than or equal to VvC. When programming memory cells to data state D, the system will test whether those memory cells have a threshold voltage greater than or equal to VvD. When programming memory cells to data state E, the system will test whether those memory cells have a threshold voltage greater than or equal to VvE. When programming memory cells to data state F, the system will test whether those memory cells have a threshold voltage greater than or equal to VvF. When programming memory cells to data state G, the system will test whether those memory cells have a threshold voltage greater than or equal to VvG. FIG. **5**C also shows Vev, which is an erase verify reference voltage to test whether a memory cell has been properly erased.

[0098] In an embodiment that utilizes full sequence programming, memory cells can be programmed from the erased data state Er directly to any of the programmed data states A-G using the process of FIG. **6** (discussed below). For example, a population of memory cells to be programmed may first be erased so that all memory cells in the population are in erased data state Er. Then, a programming process is used to program memory cells directly into data states A, B, C, D, E, F, and/or G. For example, while some memory cells are being programmed from data state Er to data state A, other memory cells are being programmed from data state Er to data state B and/or from data state Er to data state C, and so on. The arrows of FIG. **5**C represent the full sequence programming. In some embodiments, data states A-G can overlap, with control die **211** and/or memory controller **120** relying on error correction to identify the correct data being stored. Note that in some embodiments, rather than using full sequence programming, the system can use multi-pass programming processes known in the art.

[0099] In general, during verify operations and read operations, the selected word line is connected to a voltage (one example of a reference signal), a level of which is specified for each read operation (e.g., see read compare voltages/levels VrA, VrB, VrC, VrD, VrE, VrF, and VrG, of FIG. **5**C) or verify operation (e.g. see verify target voltages/levels VvA, VvB, VvC, VvD, VvE, VvF, and VvG of FIG. **5**C) in order to determine whether a threshold voltage of the concerned memory cell has reached such level. After applying the word line voltage, the conduction current of the memory cell is measured to determine whether the memory cell turned on (conducted current) in response to the voltage applied to the word line. If the conduction current is measured to be greater than a certain value, then it is assumed that the memory cell turned on and the voltage applied to the word line is greater than the threshold voltage of the memory cell. If the conduction current is not measured to be greater than the certain value, then it is assumed that the memory cell did not turn on and the voltage applied to the word line is not greater than the threshold voltage of the memory cell. During a read or verify process, the unselected memory cells are provided with one or more read pass voltages (also referred to as bypass voltages) at their control gates so that these memory cells will operate as pass gates (e.g., conducting current regardless of whether they are programmed or erased).

[0100] There are many ways to measure the conduction current of a memory cell during a read or verify operation. In one example, the conduction current of a memory cell is measured by the rate it discharges or charges a dedicated capacitor in the sense amplifier. In another example, the conduction current of the selected memory cell allows (or fails to allow) the NAND string that includes the memory cell to discharge a corresponding bit line. The voltage on the bit line is measured after a period of time to see whether it has been discharged or not. Note that the technology described herein can be used with different methods known in the art for verifying/reading. Other read and verify techniques known in the art can also be used.

[0101] FIG. **5**D depicts threshold voltage distributions when each memory cell stores four bits of data, which is another example of MLC data. FIG. **5**D depicts that there may be some overlap between the threshold voltage distributions (data states) S**0**-S**15**. The overlap may occur due to factors such as memory cells losing charge (and hence dropping in threshold voltage). Program disturb can unintentionally increase the threshold voltage of a memory cell. Likewise, read disturb can unintentionally increase the threshold voltage of a memory cell. Over time, the locations of the threshold voltage distributions may change. Such changes can increase the bit error rate, thereby increasing decoding time or even making decoding impossible. Changing the read reference voltages can help to mitigate such effects. Using ECC during the read process can fix errors and ambiguities. Note that in some embodiments, the threshold voltage distributions for a population of memory cells storing four bits of data per memory cell do not overlap and are separated from each other. The threshold voltage distributions of FIG. **5**D will include read reference voltages and verify reference voltages, as discussed above.

[0102] When using four bits per memory cell, the memory can be programmed using the full sequence programming discussed above, or multi-pass programming processes known in the art. Each threshold voltage distribution (data state) of FIG. **5**D corresponds to predetermined values for the set of data bits. The specific relationship between the data programmed into the memory cell and the threshold voltage levels of the cell depends upon the data encoding scheme adopted for the cells. Table 3 provides an example of an encoding scheme for embodiments in which each bit of data of the four bits of data stored in a memory cell are in different logical pages, referred to as a lower page (LP), middle page (MP), an upper page (UP) and top page (TP).

TABLE-US-00003 TABLE 3 S0 S1 S2 S3 S4 S5 S6 S7 S8 S9 S10 S11 S12 S13 S14 S15 TP 1 1 1 1 1 0 0 0 0 0 1 1 0 0 0 1 UP 1 1 0 0 0 0 0 0 1 1 1 1 1 1 0 0 MP 1 1 1 0 0 0 0 1 1 0 0 0 0 1 1 1 LP 1 0 0 0 1 1 0 0 0 0 0 1 1 1 1 1

[0103] FIG. **6** is a flowchart describing one embodiment of a process for programming memory cells. For purposes of this document, the term program and programming are synonymous with write and writing. In one example embodiment, the process of FIG. **6** is performed for memory array **202** using the one or more control circuits (e.g., system control logic **260**, column control circuitry **210**, row control circuitry **220**) discussed above. In one example embodiment, the process of FIG. **6** is performed by integrated memory assembly **207** using the one or more control circuits (e.g., system control logic **260**, column control circuitry **210**, row control circuitry **220**) of control die **211** to program memory cells on memory die **201**. The process includes multiple loops, each of which includes a program phase and a verify phase. The process of FIG. **6** is performed to implement the full sequence programming, as well as other programming schemes including multi-stage programming. When implementing multi-stage programming, the process of FIG. **6** is used to implement any/each stage of the multi-stage programming process.

[0104] Typically, the program voltage applied to the control gates (via a selected data word line) during a program operation is applied as a series of program voltage pulses. Between program voltage pulses are a set of verify pulses (e.g., voltage pulses) to perform verification. In many implementations, the magnitude of the program voltage pulses is increased with each successive pulse by a predetermined step size. In step **602** of FIG. **6**, the programming voltage signal (Vpgm) is initialized to the starting magnitude (e.g., ˜12-16V or another suitable level) and a program counter PC maintained by state machine **262** is initialized at **1**. In one embodiment, the group of memory cells selected to be programmed (referred to herein as the selected memory cells) are programmed concurrently and are all connected to the same word line (the selected word line). There will likely be other memory cells that are not selected for programming (unselected memory cells) that are also connected to the selected word line. That is, the selected word line will also be connected to memory cells that are supposed to be inhibited from programming. Additionally, as memory cells reach their intended target data state, they will be inhibited from further programming. Those NAND strings (e.g., unselected NAND strings) that include memory cells connected to the selected word line that are to be inhibited from programming have their channels boosted to inhibit programming. When a channel has a boosted voltage, the voltage differential between the channel and the word line is not large enough to cause programming. To assist in the boosting, in step **604** the control die will pre-charge channels of NAND strings that include memory cells connected to the selected word line that are to be inhibited from programming. In step **606**, NAND strings that include memory cells connected to the selected word line that are to be inhibited from programming have their channels boosted to inhibit programming. Such NAND strings are referred to herein as “unselected NAND strings.” In one embodiment, the unselected word lines receive one or more boosting voltages (e.g., ˜7-11 volts) to perform boosting schemes. A program inhibit voltage is applied to the bit lines coupled the unselected NAND string.

[0105] In step **608**, a program voltage pulse of the programming voltage signal Vpgm is applied to the selected word line (the word line selected for programming). If a memory cell on a NAND string should be programmed, then the corresponding bit line is biased at a program enable voltage. In step **608**, the program pulse is concurrently applied to all memory cells connected to the selected word line so that all of the memory cells connected to the selected word line are programmed concurrently (unless they are inhibited from programming). That is, they are programmed at the same time or during overlapping times (both of which are considered concurrent). In this manner all of the memory cells connected to the selected word line will concurrently have their threshold voltage change, unless they are inhibited from programming.

[0106] In step **610**, program-verify is performed, which includes testing whether memory cells being programmed have successfully reached their target data state. Memory cells that have reached their target states are locked out from further programming by the control die. Step **610** includes performing verification of programming by sensing at one or more verify reference levels. In one embodiment, the verification process is performed by testing whether the threshold voltages of the memory cells selected for programming have reached the appropriate verify reference voltage. In step **610**, a memory cell may be locked out after the memory cell has been verified (by a test of the Vt) that the memory cell has reached its target state.

[0107] In one embodiment of step **610**, a smart verify technique is used such that the system only verifies a subset of data states during a program loop (steps **604**-**628**). For example, the first program loop includes verifying for data state A (see FIG. **5**C), depending on the result of the verify operation the second program loop may perform verify for data states A and B, depending on the result of the verify operation the third program loop may perform verify for data states B and C, and so on.

[0108] In step **616**, the number of memory cells that have not yet reached their respective target threshold voltage distribution are counted. That is, the number of memory cells that have, so far, failed to reach their target state are counted. This counting can be done by state machine **262**, memory controller **120**, or another circuit. In one embodiment, there is one total count, which reflects the total number of memory cells currently being programmed that have failed the last verify step. In another embodiment, separate counts are kept for each data state.

[0109] In step **617**, the system determines whether the verify operation in the latest performance of step **610** included verifying for the last data state (e.g., data state G of FIG. **5**C). If so, then in step **618**, it is determined whether the count from step **616** is less than or equal to a predetermined limit. In one embodiment, the predetermined limit is the number of bits that can be corrected by error correction codes (ECC) during a read process for the page of memory cells. If the number of failed cells is less than or equal to the predetermined limit, then the programming process can stop and a status of “PASS” is reported in step **614**. In this situation, enough memory cells programmed correctly such that the few remaining memory cells that have not been completely programmed can be corrected using ECC during the read process. In some embodiments, the predetermined limit used in step **618** is below the number of bits that can be corrected by error correction codes (ECC) during a read process to allow for future/additional errors. When programming less than all of the memory cells for a page, the predetermined limit can be a portion (pro-rata or not pro-rata) of the number of bits that can be corrected by ECC during a read process for the page of memory cells. In some embodiments, the limit is not predetermined. Instead, it changes based on the number of errors already counted for the page, the number of program-erase cycles performed or other criteria.

[0110] If in step **617** it was determined that the verify operation in the latest performance of step **610** did not include verifying for the last data state or in step **618** it was determined that the number of failed memory cells is not less than the predetermined limit, then in step **619** the data states that will be verified in the next performance of step **610** (in the next program loop) is adjusted as per the smart verify scheme discussed above. In step **620**, the program counter PC is checked against the program limit value (PL). Examples of program limit values include 6, 12, 16, 19, 20 and 30; however, other values can be used. If the program counter PC is not less than the program limit value PL, then the program process is considered to have failed and a status of FAIL is reported in step **624**. If the program counter PC is less than the program limit value PL, then the process continues at step **626** during which time the Program Counter PC is incremented by 1 and the programming voltage signal Vpgm is stepped up to the next magnitude. For example, the next pulse will have a magnitude greater than the previous pulse by a step size ΔVpgm (e.g., a step size of 0.1-1.0 volts). After step **626**, the process continues at step **604** and another program pulse is applied to the selected word line (by the control die) so that another program loop (steps **604-626**) of the programming process of FIG. **6** is performed.

[0111] In one embodiment memory cells are erased prior to programming. Erasing is the process of changing the threshold voltage of one or more memory cells from a programmed data state to an erased data state. For example, changing the threshold voltage of one or more memory cells from state P to state E of FIG. **5**A, from states A/B/C to state E of FIG. **5**B, from states A-G to state Er of FIG. **5**C or from states S**1**-S**15** to state SO of FIG. **5**D. In one embodiment, the control circuit is configured to program memory cells in the direction from the erased data state toward the highest data state (e.g., from data state Er to data state G) and erase memory cells in the direction from the highest data state toward the erased data state (e.g., from data state G to data state Er).

[0112] One technique to erase memory cells in some memory devices is to bias a p-well (or other types of) substrate to a high voltage to charge up a NAND channel. An erase enable voltage (e.g., a low voltage) is applied to control gates of memory cells while the NAND channel is at a high voltage to erase the memory cells. Herein, this is referred to as p-well erase.

[0113] Another approach to erasing memory cells is to generate gate induced drain leakage (“GIDL”) current to charge up the NAND string channel. An erase enable voltage is applied to control gates of the memory cells, while maintaining the NAND string channel potential to erase the memory cells. Herein, this is referred to as GIDL erase. Both p-well erase and GIDL erase may be used to lower the threshold voltage (Vt) of memory cells.

[0114] In one embodiment, the GIDL current is generated by causing a drain-to-gate voltage at a GIDL generation transistor (e.g., transistors connected to SGDT**0**, SGDT**1**, SGSB**0**, and SGSB**1**). In some embodiments, a select gate (e.g., SGD or SGS) can be used as a GIDL generation transistor. A transistor drain-to-gate voltage that generates a GIDL current is referred to herein as a GIDL voltage. The GIDL current may result when the GIDL generation transistor drain voltage is significantly higher than the GIDL generation transistor control gate voltage. GIDL current is a result of carrier generation, i.e., electron-hole pair generation due to band-to-band tunneling and/or trap-assisted generation. In one embodiment, GIDL current may result in one type of carriers (also referred to a charge carriers), e.g., holes, predominantly moving into the NAND channel, thereby raising or changing the potential of the channel. The other type of carriers, e.g., electrons, are extracted from the channel, in the direction of a bit line or in the direction of a source line by an electric field. During erase, the holes may tunnel from the channel to a charge storage region of the memory cells (e.g., to charge trapping layer **493**) and recombine with electrons there, to lower the threshold voltage of the memory cells.

[0115] The GIDL current may be generated at either end (or both ends) of the NAND string. A first GIDL voltage may be created between two terminals of a GIDL generation transistor (e.g., connected to SGDT**0**, SGDT**1**) that is connected to or near a bit line to generate a first GIDL current. A second GIDL voltage may be created between two terminals of a GIDL generation transistor (e.g., SGSB**0**, SGSB**1**) that is connected to or near a source line to generate a second GIDL current. Erasing based on GIDL current at only one end of the NAND string is referred to as a one-sided GIDL erase. Erasing based on GIDL current at both ends of the NAND string is referred to as a two-sided GIDL erase. The technology described herein can be used with one-sided GIDL erase and two-sided GIDL erase.

[0116] FIG. **7** depicts the programming signal Vpgm as a series of program voltage pulses, such that one pulse of the programming signal Vpgm is applied at each performance of step **608** of FIG. **6**. These program voltage pulses are one example of doses of programming applied to a plurality of non-volatile memory cells being programmed. In one embodiment, the program voltage pulses increase in voltage magnitude from pulse-to-pulse by a step size ΔVpgm. In some embodiments, ΔVpgm can change during a programming process. As described by FIG. **6**, the system performs program-verification between the doses of programming (between or after program voltage pulses), as depicted in FIGS. **8**A and **8**B. FIG. **8**A, which illustrates an example in which program-verify is performed for one verify level, depicts two of the program voltage pulses **702** and **704** of FIG. **7**. Between program voltage pulses **702** and **704** is verify voltage pulse **710**. In one embodiment, verify voltage pulse **710** has a magnitude of any of the verify reference voltages VvA, VvB, VvC, VvD, VvE, VvF, and VvG (see FIG. **5**C) and represents the system performing program-verify (step **610**) between the doses of programming (successive iterations of step **608**). In some embodiments, between program voltage pulses the system will perform program-verify for multiple or all data states, while in other embodiments the system will perform program-verify for one data state at a time or a subset of data states. FIG. **8**B, which illustrates an example in which program-verify is performed for two verify levels, depicts two of the program voltage pulses **702** and **704** of FIG. **7**. Between program voltage pulses **702** and **704** are verify voltage pulses **710** and **712**. In one embodiment, verify voltage pulses **710** and **712** are for different data states.

[0117] In one embodiment, the system will perform program-verify at any given time (e.g., between two adjacent pair of program voltage pulses or other doses of programming) for a subset of data states. For example, when programming starts, the system will only perform program-verify for data state A (see e.g., FIG. **5**C). After at least a first amount of memory cells have successfully reached VvA, the verify reference voltage for data state A (see FIG. **5**C), then the system will only perform program-verify for data state A and data state B. After at least a first amount of memory cells have successfully reached VvB, the verify reference voltage for data state B (see FIG. **5**C), then the system will only perform program-verify for data state A, data state B and data state C. The system will stop performing program verify for data state A when all (or sufficient number of) memory cells targetd to data state A have successfully verified for data state A. And so on.

[0118] In some embodiments, the unit of programming is a page. In some embodiments, a page stores one bit of data in each memory cell connected to a same word line in a same region (e.g., regions **430**, **440**, **450**, **460** and **470** of FIG. **4**B). Therefore, when programming a large amount of data across multiple pages, the process of FIG. **6** is performed at least once for each page. For example, first the process of FIG. **6** is performed for memory cells of region **430**, followed by performing the process of FIG. **6** for memory cells of region **440**, followed by performing the process of FIG. **6** for memory cells of region **450**, followed by performing the process of FIG. **6** for memory cells of region **460**, and followed by performing the process of FIG. **6** for memory cells of region **470**. Each time the process of FIG. **6** is performed for memory cells of a region, the waveform of FIG. **7** (e.g., set of program voltage pulses that increase in voltage magnitude pulse-to-pule by a step size ΔVpgm) is applied to the word line selected for programming.

[0119] It has been observed that the memory cells being programmed do not start significantly changing threshold voltage until after a few program voltage pulses, that is until after the voltage magnitude of the program voltage pulses reaches a particular minimum voltage. However, that particular minimum voltage can change over time and can be different for different word lines, different blocks, different planes, etc. Therefore, one set of embodiments includes starting the programming process for a first set of memory cells using an initial magnitude of a programming signal (the waveform of FIG. **7**) that is lower than the possible values for that particular minimum voltage at which the first set of memory cells being programmed do start significantly changing threshold voltage, determining at which voltage magnitude of the programming signal (e.g., the program voltage pulses of FIG. **7**) the first set of memory cells being programmed do start significantly changing threshold voltage and using that determined voltage magnitude of the programming signal as the initial voltage magnitude for programming signal (e.g., the program voltage pulses of FIG. **7**) for a second set of memory cells.

[0120] In one embodiment, first the process of FIG. **6** is performed using the program voltage pulses of FIG. **7** to program memory cells of region **430** that are connected to a word line WLn, and it is observed at which voltage magnitude of the program voltage pulses of FIG. **7** that the memory cells of region **430** being programmed do start significantly changing threshold voltage (e.g., referred to as the observed voltage). Subsequently, the process of FIG. **6** is performed to program memory cells of region **440** that are connected to a word line WLn using a set of program voltage pulses that increase in voltage magnitude pulse-to-pule and have an initial voltage magnitude equal to the voltage magnitude of the observed voltage. This is followed by performing the process of FIG. **6** for memory cells of region **440** that are connected to a word line WLn using a set of program voltage pulses that increase in voltage magnitude pulse-to-pule and have an initial voltage magnitude equal to the voltage magnitude of the observed voltage, followed by performing the process of FIG. **6** for memory cells of region **450** that are connected to a word line WLn using a set of program voltage pulses that increase in voltage magnitude pulse-to-pule and have an initial voltage magnitude equal to the voltage magnitude of the observed voltage, followed by performing the process of FIG. **6** for memory cells of region **460** that are connected to a word line WLn using a set of program voltage pulses that increase in voltage magnitude pulse-to-pule and have an initial voltage magnitude equal to the voltage magnitude of the observed voltage, and followed by performing the process of FIG. **6** for memory cells of region **470** that are connected to a word line WLn using a set of program voltage pulses that increase in voltage magnitude pulse-to-pule and have an initial voltage magnitude equal to the voltage magnitude of the observed voltage.

[0121] In one example depicted in FIGS. **7**A and **7**B, when programming the memory cells of region **430** that are connected to a word line WLn, the program voltage pulses start with an initial program voltage of 8 volts and have a step size ΔVpgm of ˜0.25 volts as depicted in FIG. **7**A; and it is observed that the memory cells of region **430** being programmed start significantly changing threshold voltage at the eighth program voltage pulse (e.g., ˜10 v); therefore, when programming the memory cells of regions **440-470** that are connected to a word line WLn, the set of program voltage pulses applied to the memory cells start with an initial program voltage pulse of 10 volts as depicted in FIG. **7**B. That is, the program voltage pulses of FIG. **7**A are applied to and used to program the memory cells of region **430** that are connected to word line WLn, while the program voltage pulses of FIG. **7**B are applied to and used to program the memory cells of regions **440-470** that are connected to word line WLn. Using the program voltage pulses of FIG. **7**B for the memory cells of regions **440-470** is faster (i.e. faster programming) than using the program voltage pulses of FIG. **7**A for the memory cells of regions **440-470**.

[0122] One embodiment of the above-described process for adjusting the initial voltage magnitude of a set of program voltage pulses (i.e. the voltage magnitude of the first program voltage pulse) for programming a second set of memory cells based on programming performance of a first set of memory cells is described in more detail with respect to FIGS. **9** and **10**. FIG. **9** depicts threshold voltage distributions **902**, **904**, **906**, and **908**. Threshold voltage distributions **902** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) before any program voltage pulses are applied. In this example, the system determines the initial voltage magnitude of a set of program voltage pulses (i.e. the voltage magnitude of the first program voltage pulse) for programming a second set of memory cells based on when more than a minimum number of memory cells of the first set of memory cells being programmed have threshold voltages greater than a first threshold voltage (VL).

[0123] Threshold voltage distribution **904** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after one program voltage pulse has been applied. As can be seen, none of the memory cells of threshold voltage distributions **904** have a threshold voltage greater than the first threshold voltage (VL).

[0124] Threshold voltage distribution **906** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after two program voltage pulses have been applied. As can be seen, none of the memory cells of threshold voltage distributions **904** have a threshold voltage greater than the first threshold voltage (VL).

[0125] Threshold voltage distribution **908** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after three program voltage pulses have been applied. As can be seen, some of the memory cells of threshold voltage distribution **908** have a threshold voltage greater than the first threshold voltage (VL). The region **910** represents the memory cells having a threshold voltage greater than the first threshold voltage (VL). In this example, after the third program voltage pulse, then system will count the number of memory cells having a threshold voltage greater than VL and if the number of memory cells having a threshold voltage greater than VL is greater than X (a predetermined minimum number), then the voltage magnitude of the third program voltage pulse will be used as the initial magnitude of future sets of program voltage pulses.

[0126] FIG. **10** depicts threshold voltage distributions **1002**, **1004**, **1006**, and **1008**. Threshold voltage distributions **1002** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) before any program voltage pulses are applied. In this example, the system determines the initial voltage magnitude of a set of program voltage pulses (i.e. the voltage magnitude of the first program voltage pulse) for programming a second set of memory cells based on when more than a minimum number of memory cells of the first set of memory cells being programmed have threshold voltages greater than a first threshold voltage (VL).

[0127] Threshold voltage distributions **1004** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after one program voltage pulses has been applied. As can be seen, none of the memory cells of threshold voltage distributions **1004** have a threshold voltage greater than the first threshold voltage (VL).

[0128] Threshold voltage distributions **1006** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after two program voltage pulses have been applied. As can be seen, none of the memory cells of threshold voltage distributions **1004** have a threshold voltage greater than the first threshold voltage (VL).

[0129] Threshold voltage distributions **1008** represents the first group of memory cells being programmed (e.g., the memory cells of region **430**) after three program voltage pulses have been applied. As can be seen, some of the memory cells of threshold voltage distributions **1008** have a threshold voltage greater than the first threshold voltage (VL). The region **1010** represents the memory cells having a threshold voltage greater than the first threshold voltage (VL) and less thana second threshold voltage (VH). The region **1012** represents the memory cells having a threshold voltage greater than the second threshold voltage (VH). In this example, after the third program voltage pulse, then system will count the number of memory cells having a threshold voltage greater than VL and the number of memory cells having a threshold voltage greater than VH. If the number of memory cells having a threshold voltage greater than VL is greater than X (a first predetermined minimum number) and the number of memory cells having a threshold voltage greater than VH is not greater than Y (a second predetermined minimum number), then the voltage magnitude of the third program voltage pulse will be used as the initial magnitude for future sets of program voltage pulses.

[0130] If the number of memory cells having a threshold voltage greater than VL is greater than X and the number of memory cells having a threshold voltage greater than VH is greater than Y, then the voltage magnitude of the third program voltage pulse less an offset will be used as the initial magnitude for future sets of program voltage pulses. Examples of the offset can be the step size ΔVpgm, a fraction of the step size (e.g., half of the step size), another fixed amount or a variable amount based on operating conditions.

[0131] In some embodiments, the process of for adjusting the initial voltage magnitude of a set of program voltage pulses (i.e. the voltage magnitude of the first program voltage pulse) for programming a second set of memory cells based on programming performance of a first set of memory cells is performed once and then the new initial magnitude is used thereafter. In other embodiments, the process of for adjusting the initial voltage magnitude of a set of program voltage pulses for programming a second set of memory cells based on programming performance of a first set of memory cells is performed every programming process, periodically, only after a predetermined number of read errors, only after many program/erase cycles, etc.

[0132] FIG. **11** is a flow chart describing one embodiment of a process for performing a verify process that sets the initial voltage magnitude of a future set of program voltage pulses. That is, the process of FIG. **11** is an example implementation of step **610** in order to implement the above-described technology for adjusting the initial voltage magnitude of a set of program voltage pulses for programming a second set of memory cells based on programming performance of a first set of memory cells. In some example implementations, the process of FIG. **11** can be performed by any one of the one or more control circuits discussed above. The process of FIG. **11** can be performed entirely by a control circuit on memory die **200** (see FIG. **2**A) or entirely by a control circuit on integrated memory assembly **207** (see FIG. **2**B), rather than by memory controller **120**. In one example, the process of

[0133] FIG. **11** is performed by or at the direction of state machine **262**, using other components of System Control Logic **260**, Column Control Circuitry **210** and Row Control Circuitry **220**. In another embodiment, the process of FIG. **11** is performed by memory controller **120** in combination with System Control Logic **260**, Column Control Circuitry **210** and Row Control Circuitry **220**. In some embodiments, the process of FIG. **11** is performed on any of the non-volatile memories discussed above.

[0134] In step **1102** of FIG. **11**, the control circuit first determines whether the initial voltage magnitude of the set of program voltage pulses for the next set of memory cells to be programmed has already been set. If so, then in step **1104** the control circuit performs program-verify at one or more appropriate verify reference voltages, as described above. In step **1106**, the control circuit locks out memory cells having successfully completed program-verify as these memory cells have reached their target state.

[0135] If, in step **1102**, the control circuit determines that the initial voltage magnitude of the set of program voltage pulses for the next set of memory cells to be programmed has not yet been set, then in step **1120** the control circuit senses whether the memory cells being programmed have threshold voltages greater than VL (see FIG. **10**). In step **1122**, the control circuit counts the number of memory cells having threshold voltages greater than VL. In step **1124** the control circuit senses whether the memory cells being programmed have threshold voltages greater than VH (see FIG. **10**). In step **1126**, the control circuit counts the number of memory cells having threshold voltages greater than VH. In step **1128**, the control circuit determines whether the number of memory cells having a threshold voltage greater than VL is greater than X (a first predetermined minimum number). If not, then the verify operation is complete (step **1130**). If the number of memory cells having a threshold voltage greater than VL is greater than X, then in step **1132** the control circuit determines whether the number of memory cells having a threshold voltage greater than VH is greater than Y (a second predetermined minimum number). If the number of memory cells having a threshold voltage greater than VH is not greater than Y, then (in step **1134**) the control circuit sets the initial voltage magnitude of the program voltage pulses to the voltage magnitude of the latest program voltage pulse that was just applied. If the number of memory cells having a threshold voltage greater than VH is greater than Y, then (in step **1136**) the control circuit sets the initial voltage magnitude of the program voltage pulses to the voltage magnitude of the latest program voltage pulse that was just applied less an offset (e.g., ½ ΔVpgm).

[0136] FIG. **12** is a timing diagram describing certain signals during one embodiment of a programming process that includes the program-verify process of FIG. **11** based on the embodiments of FIGS. **9** and/or **10**. FIG. **12** shows four signals: “sel WL”, “unsel WL”, “sel SGD” and “usel SGD.” The signal sel WL is the voltage applied to the word line (e.g., of word lines WL**0**-WL**161** of FIG. **4**C) that is selected for programming. In an embodiment with multiple regions (e.g., **430-470**), memory cells are programmed one region at a time. Therefore, memory cells connected to sel WL and in the selected region will be programmed. The signal unsel WL represents the voltage applied all (or most) word lines other than the word line selected for programming. The signal sel SGD represent the voltage applied to the drain side select gates (e.g., SGDT**0**, SGDT**1**, SGD**0** and SGD**1**) for the region (e.g., of regions **430-470**) selected for programming (see also FIG. **4**F). The signal usel SGD represent the voltage applied to the drain side select gates (e.g., SGDT**0**, SGDT**1**, SGD**0** and SGD**1**) for the regions (e.g., of regions **430-470**) that are not selected for programming (see also FIG. **4**F).

[0137] At the beginning of the time interval depicted in FIG. **12**, all signals are at Vss (e.g., ground potential or Ov). At time to, sel WL is raised to Vpgm (the magnitude of the program voltage pulse), unsel WL is raised to Vpass (e.g., ˜8 v), and sel SGD is raised to Vsgd (e.g., 1.5-3 v). At time t**1**, the program voltage pulse ends and sel WL is lowered to Vss or a small negative voltage, unsel WL is lowered to Vread (e.g., 4-6 v) and unsel SGD is raised to Vsgd. At time t**2**, unsel SGD is lowered back to Vss and sel WL is lowered to VL. FIG. **12** depicts VL to be a negative voltage. In other embodiments, VL is a small positive voltage. At time t**3**. sel; WL is raised to Vss, unsel WL is lowered to Vss, and sel SGD is lowered to Vss. Between t**2** and t**3** corresponds to step **1120** (sensing at VL). Between t**3** and t**4** corresponds to step **1122** (count memory cells with Vth>VL). At time t**4** a voltage spike is applied to sel WL, unsel WL is raised to Vread, sel SGD is raised to Vsg and usel SGD is raised to Vsg. After the voltage spike at t**4**, sel WL is lowered to VH to perform step **1124**. FIG. **12** depicts VH to be a negative voltage. In other embodiments, VH is a

small positive voltage and VH>VL. At time t**5**, sel WL is raised to Vss, unsel WL is lowered to Vss, and sel SGD is lowered to Vss. Between t**5** and t**6** corresponds to step **1126** (count memory cells with Vth>VH). At time t**6**, sel WL is raised to the voltage magnitude of the next program voltage pulse, unsel WL is raised to Vpass and sel SGD is raised to Vsg. And the process continues.
[0138] In some embodiments, the voltage spike at t**4** was used to prevent read disturb by turning on an entire NAND string to equalize the channel potential and remove stray charge that could be inserted into the charge trapping layer. However, it has been observed that the voltage spike is not necessary to prevent read disturb and can be removed. By removing the voltage spike, the time used for the programming process is shortened (i.e. performance increase).
[0139] FIG. **13** is a timing diagram describing certain signals during one embodiment of a programming process that includes the program-verify process of FIG. **11** based on the embodiments of FIGS. **9** and/or **10**. However, in the embodiment of FIG. **13** the voltage spile has been removed. That is, the control circuit applies a first test voltage (VL) to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a first condition followed by applying a second test voltage (VH) to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a second condition without applying a voltage spike on the word line connected to the first group of non-volatile memory cells in a time period between the applying the first test voltage and the applying the second test voltage. FIG. **13** shows the four signals “sel WL”, “unsel WL”, “sel SGD” and “usel SGD.” From t**0**-t**4**, FIG. **13** is the same as FIG. **12**.
[0140] At time t**4** in FIG. **13**, sel WL is lowered to VH. No voltage spike is applied. Also at time t**4**, unsel WLis raised to Vread and sel SGD is raised to Vsg. At time ta, sel WL is raised to Vss. At time tb, sel WL is raised to the voltage magnitude of the next program voltage pulse. At time tc, unsel WL is raised to Vpass and sel SGD is raised to Vsg. And the process continues. FIG. **13** depicts time period **1302**, representing the time between the applying the first test voltage VL and the applying the second test voltage VH. No voltage spike is applied to sel WL during time period **1302**. The control circuit is configured to apply and hold a ground voltage potential (e.g., Vss) to the selected word line sel WL connected to the group of non-volatile memory cells being programmed during the time period **1302** between the applying the first test voltage VL and the applying the second test voltage VH.
[0141] FIG. **14** is a flow chart describing one embodiment of a process for programming non-volatile memory cells that includes adjusting the initial voltage magnitude of a set of program voltage pulses for programming a second set of memory cells based on programming performance of a first set of memory cells. In some example implementations, the process of FIG. **14** can be performed by any one of the one or more control circuits discussed above. The process of FIG. **14** can be performed entirely by a control circuit on memory die **200** (see FIG. **2**A) or entirely by a control circuit on integrated memory assembly **207** (see FIG. **2**B), rather than by memory controller **120**. In one example, the process of FIG. **14** is performed by or at the direction of state machine **262**, using other components of System Control Logic **260**, Column Control Circuitry **210** and Row Control Circuitry **220**. In another embodiment, the process of FIG. **14** is performed by memory controller **120** in combination with System Control Logic **260**, Column Control Circuitry **210** and Row Control Circuitry **220**. In some embodiments, the process of FIG. **14** is performed on any of the non-volatile memories discussed above.
[0142] In step **1402**, the control circuit applies a first set of doses of programming (e.g., see FIG. **7**A) to a word line connected to the first group of non-volatile memory cells in order to program the first group of non-volatile memory cells. The first set of doses of programming has a first initial voltage magnitude (see 8 v of FIG. **7**A).
[0143] In step **1404**, between an adjacent pair of doses of programming of the first set of doses: the control circuit applies a first test voltage (e.g., VL) to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a first

condition (e.g., **1010**) followed by applying a second test voltage (e.g., VH) to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a second condition (e.g., **1012**) without applying a voltage spike on the word line connected to the first group of non-volatile memory cells in a time period (e.g., **1302**) between the applying the first test voltage and the applying the second test voltage.

[0144] In step **1406**, the control circuit applies a second set of doses of programming (see e.g., FIG. **7**B) to a word line connected to the second group of non-volatile memory cells in order to program the second group of non-volatile memory cells. The second set of doses of programming includes a second initial voltage magnitude (see 10 v of FIG. **7**B). In one embodiment, the first group of non-volatile memory cells are in region **430** and the second group of non-volatile memory cells are region **440**, with the first group of non-volatile memory cells and the second group of non-volatile memory cells all connected to the same word line. In other embodiments, the two groups of memory cells are on different word lines in the same region, different word lines in the same block or other arrangements.

[0145] In step **1408**, prior to applying the second set of doses of programming, the control circuit sets the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition. In one embodiment, setting the second initial voltage magnitude comprises setting the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming in response to more than a first number of memory cells being in the first condition and less than a second number of memory cells being in the second condition (step **1420**) and setting the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming less an offset (e.g., ΔVpgm or ½ ΔVpgm) in response to more than a first number of memory cells being in the first condition and more than a second number of memory cells being in the second condition (step **1422**).

[0146] In one example implementation, the first group of non-volatile memory cells are connected to a first set of select gates; the first set of select gates are connected to a first selection line (e.g., SGDT**0**, SGDT**1**, SGD**0** and SGD**1** for the region of regions **430-470** that the first group of non-volatile memory cells are positioned in); the second group of non-volatile memory cells are connected to a second set of select gates; the second set of select gates are connected to a second selection line (e.g., SGDT**0**, SGDT**1**, SGD**0** and SGD**1** for the region of regions **430-470** that the second group of non-volatile memory cells are positioned in); the control circuit is further configured to apply a first voltage pulse (e.g., at Vsg-see FIG. **13**) to the first selection line while applying the first test voltage to the word line connected to the first group of non-volatile memory cells and to apply a second voltage pulse (e.g., at Vsg-see FIG. **13**) to the first selection line (e.g., sel SGD) while applying the second test voltage to the word line connected to the first group of non-volatile memory cells; and the control circuit is further configured to apply a third voltage pulse to the second selection (e.g., usel SGD) line during the first voltage pulse and to apply a ground voltage to the second selection line during the second voltage pulse.

[0147] FIG. **15** is a flow chart describing one embodiment of a process for programming non-volatile memory cells that includes adjusting the initial voltage magnitude of a set of program voltage pulses for programming a second set of memory cells based on programming performance of a first set of memory cells. In some example implementations, the process of FIG. **15** can be performed by any one of the one or more control circuits discussed above. The process of FIG. **15** can be performed entirely by a control circuit on memory die **200** (see FIG. **2**A) or entirely by a control circuit on integrated memory assembly **207** (see FIG. **2**B), rather than by memory controller **120**. In one example, the process of FIG. **15** is performed by or at the direction of state machine **262**, using other components of System Control Logic **260**, Column Control Circuitry **210** and Row Control Circuitry **220**. In another embodiment, the process of FIG. **15** is performed by memory controller **120** in combination with System Control Logic **260**, Column Control Circuitry

210 and Row Control Circuitry **220**. In some embodiments, the process of FIG. **15** is performed on any of the non-volatile memories discussed above.

[0148] Step **1502** includes the control circuit performing a first programming process for a first group of non-volatile memory cells including applying a first set of doses of programming (see FIG. **7**A) to a word line connected to the first group of non-volatile memory cells. Step **1504** includes the control circuit performing a second programming process for a second group of non-volatile memory cells including applying a second set of doses of programming (see FIG. **7**B) to a word line connected to the second group of non-volatile memory cells, the second set of doses of programming start with an initial voltage for the second group of non-volatile memory cells. Step **1506** includes the control circuit determining the initial voltage for the second group of non-volatile memory cells based on the first programming process for the first group of non-volatile memory cells (see e.g., FIG. **10**).

[0149] In one embodiment, the determining the initial voltage for the second group of non-volatile memory cells of step **1506** comprises: after an individual dose of the first set of doses of programming and prior to a next dose of the first set of doses of programming, performing a first verify operation (see e.g., step **1120**) for the first group of non-volatile memory cells to determine whether the first group of non-volatile memory cells have reached a first condition (e.g., **1010**) in step **1520**; counting the number of non-volatile memory cells of the first group that are in the first condition in step **1522**; after the individual dose of the first set of doses of programming and prior to the next dose of the first set of doses of programming, performing a second verify operation (see e.g., step **1124**) for the first group of non-volatile memory cells without applying a voltage spike to the word line connected to the first group of non-volatile memory cells after the first verify operation and before the second verify operation in step **1524**; counting the number of non-volatile memory cells of the first group that are in the second condition (e.g., **1012**) in step **1526**; and in response to the number of non-volatile memory cells of the first group that are counted to be in the first condition being greater than a first amount, setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming if the number of non-volatile memory cells of the first group that are counted to be in the second condition is not greater than a second amount and setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming less a voltage magnitude if the number of non-volatile memory cells of the first group that are counted to be in the second condition is greater than the second amount in step **1528**.

[0150] In one set of embodiments, the word line connected to the first group of non-volatile memory cells is the same word line as the word line connected to the second group of non-volatile memory cells. In another set of embodiments, the word line connected to the first group of non-volatile memory cells is a different word line as the word line connected to the second group of non-volatile memory cells and the first group of non-volatile memory cells and the second group of non-volatile memory cells are in a same block. In one example implementation: the first set of doses of programming comprise a first set of program voltage pulses; the individual dose of the first set of doses is an individual program voltage pulse; and the second set of doses of programming comprise a second set of program voltage pulses. In one embodiment, the first condition is a threshold voltage greater than a first threshold voltage VL and the second condition is a threshold voltage greater than a second threshold voltage VH.

[0151] A non-volatile memory has been described that determines an initial magnitude of a programming signal for a second set of memory cells based on testing during the programming of the first set of memory cells without applying a voltage spike between the sensing operations in order to increase performance (e.g., speed of programming) and decrease power usage (i.e. due to the omission of the voltage spike).

[0152] One embodiment includes a non-volatile storage apparatus, comprising: a non-volatile memory comprising a plurality of non-volatile memory cells and word lines connected to the

plurality of non-volatile memory cells, the plurality of non-volatile memory cells include a first group of non-volatile memory cells and a second group of non-volatile memory cells; and a control circuit connected to the plurality of non-volatile memory cells. The control circuit is configured to: apply a first set of doses of programming to a word line connected to the first group of non-volatile memory cells in order to program the first group of non-volatile memory cells, the first set of doses of programming has a first initial voltage magnitude, between an adjacent pair of doses of programming of the first set of doses: apply a first test voltage to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a first condition followed by applying a second test voltage to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a second condition without applying a voltage spike on the word line connected to the first group of non-volatile memory cells in a time period between the applying the first test voltage and the applying the second test voltage, apply a second set of doses of programming to a word line connected to the second group of non-volatile memory cells in order to program the second group of non-volatile memory cells, the second set of doses of programming includes a second initial voltage magnitude, and prior to applying the second set of doses of programming, set the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition.
[0153] In one example implementation, the first set of doses of programming comprise a first set of program voltage pulses; and the second set of doses of programming comprise a second set of program voltage pulses.
[0154] In one example implementation, the word line connected to the first group of non-volatile memory cells is the same word line as the word line connected to the second group of non-volatile memory cells.
[0155] In one example implementation, word line connected to the first group of non-volatile memory cells is the different word line then the word line connected to the second group of non-volatile memory cells.
[0156] In one example implementation, the first group of non-volatile memory cells are connected to a first set of select gates; the first set of select gates are connected to a first selection line; the second group of non-volatile memory cells are connected to a second set of select gates; the second set of select gates are connected to a second selection line; the control circuit is further configured to apply a first voltage pulse to the first selection line while applying the first test voltage to the word line connected to the first group of non-volatile memory cells and to apply a second voltage pulse to the first selection line while applying the second test voltage to the word line connected to the first group of non-volatile memory cells; and the control circuit is further configured to apply a third voltage pulse to the second selection line during the first voltage pulse and to apply a ground voltage to the second selection line during the second voltage pulse.
[0157] In one example implementation, the control circuit is configured to set the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming in response to more than a first number of memory cells being in the first condition.
[0158] In one example implementation, the control circuit is configured to set the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming in response to more than a first number of memory cells being in the first condition and less than a second number of memory cells being in the second condition.
[0159] In one example implementation, the control circuit is configured to set the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming less an offset in response to more than a first number of memory cells being in the first condition and more than a second number of memory cells being in the second condition.

[0160] In one example implementation, the first condition is a threshold voltage greater than a first threshold voltage; the second condition is a threshold voltage greater than a second threshold voltage; and the offset is equal to a difference between the second threshold voltage and the first threshold voltage.
[0161] In one example implementation, the first set of doses of programming include program voltage pulses that increase pulse-to-pule by a step size; and the offset is equal to half the step size.
[0162] In one example implementation, the control circuit is configured to apply and hold a ground voltage potential to the word line connected to the first group of non-volatile memory cells during the time period between the applying the first test voltage and the applying the second test voltage.
[0163] One embodiment includes a method, comprising: performing a first programming process for a first group of non-volatile memory cells including applying a first set of doses of programming to a word line connected to the first group of non-volatile memory cells; performing a second programming process for a second group of non-volatile memory cells including applying a second set of doses of programming to a word line connected to the second group of non-volatile memory cells, the second set of doses of programming start with an initial voltage for the second group of non-volatile memory cells; and determining the initial voltage for the second group of non-volatile memory cells based on the first programming process for the first group of non-volatile memory cells. The determining the initial voltage for the second group of non-volatile memory cells comprises: after an individual dose of the first set of doses of programming and prior to a next dose of the first set of doses of programming, performing a first verify operation for the first group of non-volatile memory cells to determine whether the first group of non-volatile memory cells have reached a first condition, counting the number of non-volatile memory cells of the first group that are in the first condition, after the individual dose of the first set of doses of programming and prior to the next dose of the first set of doses of programming, performing a second verify operation for the first group of non-volatile memory cells without applying a voltage spike to the word line connected to the first group of non-volatile memory cells after the first verify operation and before the second verify operation, counting the number of non-volatile memory cells of the first group that are in the second condition, and in response to the number of non-volatile memory cells of the first group that are counted to be in the first condition being greater than a first amount, setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming if the number of non-volatile memory cells of the first group that are counted to be in the second condition is not greater than a second amount and setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming less a voltage magnitude if the number of non-volatile memory cells of the first group that are counted to be in the second condition is greater than the second amount.
[0164] In one example implementation, the word line connected to the first group of non-volatile memory cells is the same word line as the word line connected to the second group of non-volatile memory cells.
[0165] In one example implementation, the word line connected to the first group of non-volatile memory cells is a different word line as the word line connected to the second group of non-volatile memory cells; and the first group of non-volatile memory cells and the second group of non-volatile memory cells are in a same block.
[0166] In one example implementation, the first set of doses of programming comprise a first set of program voltage pulses; the individual dose of the first set of doses is an individual program voltage pulse; and the second set of doses of programming comprise a second set of program voltage pulses.
[0167] In one example implementation, the first condition is a threshold voltage greater than a first threshold voltage; and the second condition is a threshold voltage greater than a second threshold voltage.
[0168] In one example implementation, applying a first set of program voltage pulses to a first

group of non-volatile memory cells via a first word line, the first set of program voltage pulses has a first initial voltage magnitude; between adjacent program voltage pulses of the first set of program voltage pulses: applying a first test voltage to the word line to determine a number of memory cells of the first group in a first condition followed by applying a second test voltage to the word line to determine a number of memory cells of the first group in a second condition without applying a voltage spike on the word line in a time period between the applying the first test voltage and the applying the second test voltage; applying a second set of program voltage pulses to a second group of non-volatile memory cells via the first word line after applying the first set of program voltage pulses to the first group of non-volatile memory cells, the second set of program voltage pulses has a second initial voltage magnitude that is greater in voltage magnitude than the first initial voltage magnitude; and after applying the first set of program voltage pulses to the first group of non-volatile memory cells and prior to applying the second set of program voltage pulses to the second group of non-volatile memory cells, setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition.

[0169] In one example implementation, the first condition is a threshold voltage greater than a first threshold voltage; and the second condition is a threshold voltage greater than a second threshold voltage.

[0170] In one example implementation, the setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition comprises: setting the second initial voltage magnitude to a voltage magnitude of a first program voltage pulse of the adjacent program voltage pulses.

[0171] In one example implementation, the setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition comprises: setting the second initial voltage magnitude to a voltage magnitude of a first program voltage pulse of the adjacent program voltage pulses less an offset in response to more than a first number of memory cells being in the first condition and more than a second number of memory cells being in the second condition.

[0172] For purposes of this document, reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” or “another embodiment” may be used to describe different embodiments or the same embodiment.

[0173] For purposes of this document, a connection may be a direct connection or an indirect connection (e.g., via one or more other parts). In some cases, when an element is referred to as being connected or coupled to another element, the element may be directly connected to the other element or indirectly connected to the other element via one or more intervening elements. When an element is referred to as being directly connected to another element, then there are no intervening elements between the element and the other element. Two devices are “in communication” if they are directly or indirectly connected so that they can communicate electronic signals between them.

[0174] For purposes of this document, the term “based on” may be read as “based at least in part on.”

[0175] For purposes of this document, without additional context, use of numerical terms such as a “first” object, a “second” object, and a “third” object may not imply an ordering of objects, but may instead be used for identification purposes to identify different objects.

[0176] For purposes of this document, the term “set” of objects may refer to a “set” of one or more of the objects.

[0177] The foregoing detailed description has been presented for purposes of illustration and description. It is not intended to be exhaustive or to limit to the precise form disclosed. Many modifications and variations are possible in light of the above teaching. The described embodiments were chosen in order to best explain the principles of the proposed technology and its practical application, to thereby enable others skilled in the art to best utilize it in various embodiments and with various modifications as are suited to the particular use contemplated. It is intended that the scope be defined by the claims appended hereto.

Claims

1. A non-volatile storage apparatus, comprising: a non-volatile memory comprising a plurality of non-volatile memory cells and word lines connected to the plurality of non-volatile memory cells, the plurality of non-volatile memory cells include a first group of non-volatile memory cells and a second group of non-volatile memory cells; and a control circuit connected to the plurality of non-volatile memory cells, the control circuit is configured to: apply a first set of doses of programming to a word line connected to the first group of non-volatile memory cells in order to program the first group of non-volatile memory cells, the first set of doses of programming has a first initial voltage magnitude, between an adjacent pair of doses of programming of the first set of doses: apply a first test voltage to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a first condition followed by applying a second test voltage to the word line connected to the first group of non-volatile memory cells to determine a number of memory cells of the first group in a second condition without applying a voltage spike on the word line connected to the first group of non-volatile memory cells in a time period between the applying the first test voltage and the applying the second test voltage, apply a second set of doses of programming to a word line connected to the second group of non-volatile memory cells in order to program the second group of non-volatile memory cells, the second set of doses of programming includes a second initial voltage magnitude, and prior to applying the second set of doses of programming, set the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition.
2. The apparatus of claim 1, wherein: the first set of doses of programming comprise a first set of program voltage pulses; and the second set of doses of programming comprise a second set of program voltage pulses.
3. The apparatus of claim 1, wherein: the word line connected to the first group of non-volatile memory cells is the same word line as the word line connected to the second group of non-volatile memory cells.
4. The apparatus of claim 1, wherein: word line connected to the first group of non-volatile memory cells is the different word line then the word line connected to the second group of non-volatile memory cells.
5. The apparatus of claim 1, wherein: the first group of non-volatile memory cells are connected to a first set of select gates; the first set of select gates are connected to a first selection line; the second group of non-volatile memory cells are connected to a second set of select gates; the second set of select gates are connected to a second selection line; the control circuit is further configured to apply a first voltage pulse to the first selection line while applying the first test voltage to the word line connected to the first group of non-volatile memory cells and to apply a second voltage pulse to the first selection line while applying the second test voltage to the word line connected to the first group of non-volatile memory cells; and the control circuit is further configured to apply a third voltage pulse to the second selection line during the first voltage pulse and to apply a ground voltage to the second selection line during the second voltage pulse.
6. The apparatus of claim 1, wherein: the control circuit is configured to set the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming in response to more than a first number of memory cells being in the first condition.
7. The apparatus of claim 1, wherein: the control circuit is configured to set the second initial

voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming in response to more than a first number of memory cells being in the first condition and less than a second number of memory cells being in the second condition.

8. The apparatus of claim 1, wherein: the control circuit is configured to set the second initial voltage magnitude to a voltage magnitude of a first dose of programming of the adjacent pair of doses of programming less an offset in response to more than a first number of memory cells being in the first condition and more than a second number of memory cells being in the second condition.

9. The apparatus of claim 8, wherein: the first condition is a threshold voltage greater than a first threshold voltage; the second condition is a threshold voltage greater than a second threshold voltage; and the offset is equal to a difference between the second threshold voltage and the first threshold voltage.

10. The apparatus of claim 8, wherein: the first set of doses of programming include program voltage pulses that increase pulse-to-pule by a step size; and the offset is equal to half the step size.

11. The apparatus of claim 1, wherein: the control circuit is configured to apply and hold a ground voltage potential to the word line connected to the first group of non-volatile memory cells during the time period between the applying the first test voltage and the applying the second test voltage.

12. A method, comprising: performing a first programming process for a first group of non-volatile memory cells including applying a first set of doses of programming to a word line connected to the first group of non-volatile memory cells; performing a second programming process for a second group of non-volatile memory cells including applying a second set of doses of programming to a word line connected to the second group of non-volatile memory cells, the second set of doses of programming start with an initial voltage for the second group of non-volatile memory cells; and determining the initial voltage for the second group of non-volatile memory cells based on the first programming process for the first group of non-volatile memory cells, the determining the initial voltage for the second group of non-volatile memory cells comprises: after an individual dose of the first set of doses of programming and prior to a next dose of the first set of doses of programming, performing a first verify operation for the first group of non-volatile memory cells to determine whether the first group of non-volatile memory cells have reached a first condition, counting the number of non-volatile memory cells of the first group that are in the first condition, after the individual dose of the first set of doses of programming and prior to the next dose of the first set of doses of programming, performing a second verify operation for the first group of non-volatile memory cells without applying a voltage spike to the word line connected to the first group of non-volatile memory cells after the first verify operation and before the second verify operation, counting the number of non-volatile memory cells of the first group that are in the second condition, and in response to the number of non-volatile memory cells of the first group that are counted to be in the first condition being greater than a first amount, setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming if the number of non-volatile memory cells of the first group that are counted to be in the second condition is not greater than a second amount and setting the initial voltage based on voltage magnitude of the individual dose of the first set of doses of programming less a voltage magnitude if the number of non-volatile memory cells of the first group that are counted to be in the second condition is greater than the second amount.

13. The method of claim 12, wherein: the word line connected to the first group of non-volatile memory cells is the same word line as the word line connected to the second group of non-volatile memory cells.

14. The method of claim 12, wherein: the word line connected to the first group of non-volatile memory cells is a different word line as the word line connected to the second group of non-volatile memory cells; and the first group of non-volatile memory cells and the second group of non-volatile memory cells are in a same block.

15. The method of claim 12, wherein: the first set of doses of programming comprise a first set of program voltage pulses; the individual dose of the first set of doses is an individual program voltage pulse; and the second set of doses of programming comprise a second set of program voltage pulses.

16. The method of claim 14, wherein: the first condition is a threshold voltage greater than a first threshold voltage; and the second condition is a threshold voltage greater than a second threshold voltage.

17. A method, comprising: applying a first set of program voltage pulses to a first group of non-volatile memory cells via a first word line, the first set of program voltage pulses has a first initial voltage magnitude; between adjacent program voltage pulses of the first set of program voltage pulses: applying a first test voltage to the word line to determine a number of memory cells of the first group in a first condition followed by applying a second test voltage to the word line to determine a number of memory cells of the first group in a second condition without applying a voltage spike on the word line in a time period between the applying the first test voltage and the applying the second test voltage; applying a second set of program voltage pulses to a second group of non-volatile memory cells via the first word line after applying the first set of program voltage pulses to the first group of non-volatile memory cells, the second set of program voltage pulses has a second initial voltage magnitude that is greater in voltage magnitude than the first initial voltage magnitude; and after applying the first set of program voltage pulses to the first group of non-volatile memory cells and prior to applying the second set of program voltage pulses to the second group of non-volatile memory cells, setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition.

18. The method of claim 17, wherein: the first condition is a threshold voltage greater than a first threshold voltage; and the second condition is a threshold voltage greater than a second threshold voltage.

19. The method of claim 18, wherein the setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition comprises: setting the second initial voltage magnitude to a voltage magnitude of a first program voltage pulse of the adjacent program voltage pulses.

20. The method of claim 18, wherein the setting the second initial voltage magnitude based on the number of memory cells of the first group in the first condition and the number of memory cells of the first group in the second condition comprises: setting the second initial voltage magnitude to a voltage magnitude of a first program voltage pulse of the adjacent program voltage pulses less an offset in response to more than a first number of memory cells being in the first condition and more than a second number of memory cells being in the second condition.